CNC Data Management Tool Specifications

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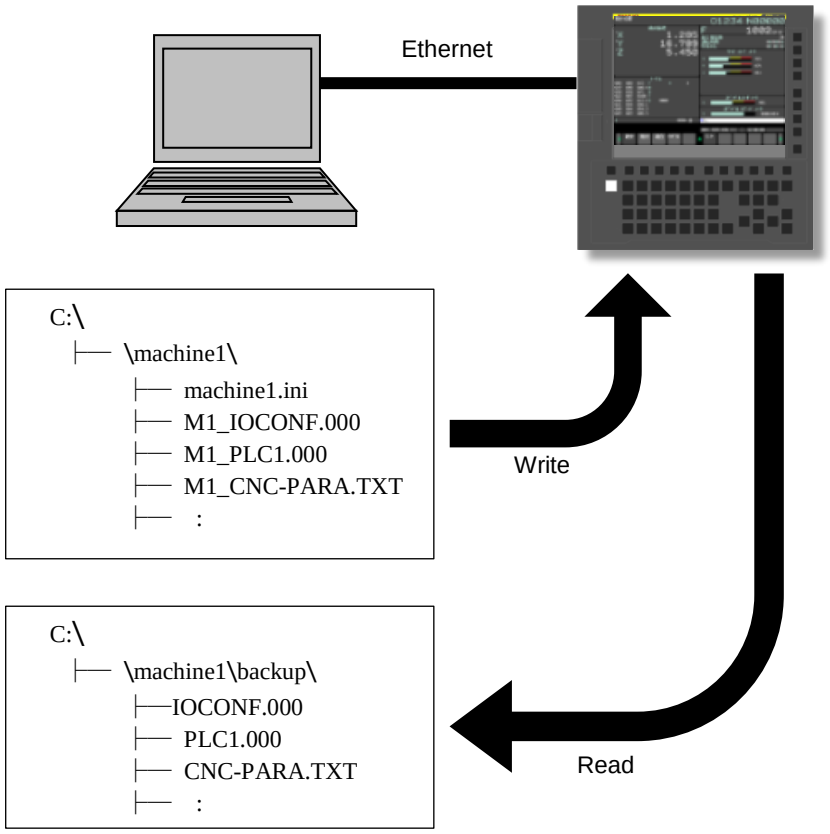
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1 CNC DATA MANAGEMENT TOOL

1.1 OVERVIEW

This tool is a PC tool for managing application files such as macro executors and text files such as parameters for each model and can easily write them to CNC or read them from CNC. <5>

By using this tool, it is possible to significantly reduce the time required for writing various data associated with the new startup of CNC and migrating various data associated with maintenance work.



<5>

⚠ WARNING

1. When writing parameters, user files, etc., be sure to fully understand the behavior before writing. If you write wrong parameters or user files may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.
2. The writing file of this tool assumes the data format of the file output from CNC. Therefore, you cannot write files with different data formats. If you write the wrong file may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.
3. This tool writes the file information to CNC sequentially from the first line. Therefore, if this tool is forcibly terminated or writing ends in error, the file information before the termination will be written to CNC. Please check the log file etc. to make sure that the target file has been written to the end. If the writing was not done correctly may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.
4. Do not operate CNC when using this tool. If a mode is changed or a reset operation is executed, writing / reading may not be executed normally.

<5>
<5>

2 ENVIRONMENT

2.1 OPERATING ENVIRONMENT

Operating PC environment

Target OS:

Microsoft® Windows® XP Professional + Service Pack 3
Microsoft® Windows® 7 Professional 32/64-bit Edition
Microsoft® Windows® 8 Professional 64-bit Edition
Microsoft® Windows® 10 Professional 64-bit Edition
Microsoft® Windows® 11 Professional 64-bit Edition <11>

Communication environment

Ethernet

Target models

FANUC Series 0*i*-D/PD/Mate-D, FANUC Series 0*i*-F/PF/F Plus/PF Plus/LF Plus, FANUC Series 30*i*/31*i*/32*i*-A/B/B Plus, <14>
FANUC Series 30*i*/31*i*-LB/PB/LB Plus, FANUC Series 31*i*-PB Plus, LSC*i*-A, FANUC Series 31*i*-WA/WB, <14>
FANUC Series 35*i*-B, Power Motion *i*-A/A Plus, FANUC Series 16*i*/18*i*/21*i*-B <12>

CAUTION
1. In this specification, program names and device names other than those of our products are listed, but these include registered trademarks of each manufacturer. However, the ® and ™ marks may not be specified in the text.
2. Microsoft, Windows are registered trademarks of Microsoft Corporation of the United States of America.

2.2 INSTALL AND UNINSTALL CNC DATA MANAGEMENT TOOL

CAUTION
The administrator's privilege in your personal computer is necessary for installing or uninstalling of CNC Data Management Tool.

2.2.1 INSTALL

This section describes how to install "CNC Data Management Tool".
(1) Insert the CD into your CD drive.
(2) Click the Start button of Windows.
(3) Click Computer or My computer.
(4) Double-click SETUP.EXE in the CD drive to start the setup.
The installer (Fig. 2.2.1(a)) will start. Follow the message and proceed with the installation.

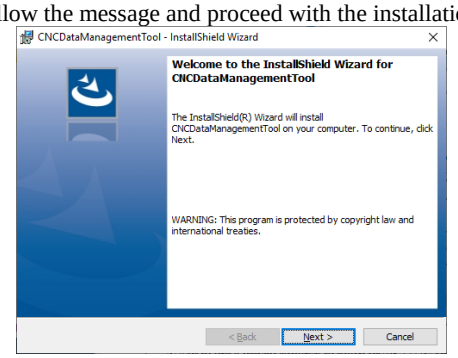


Fig. 2.2.1(a) Installation start screen

(5) When the setup is completed successfully, the message box (Fig. 2.2.1 (b)) notifying that the installation is completed is displayed. Also, a shortcut icon for CNC Data Management Tool will be displayed on the desktop.

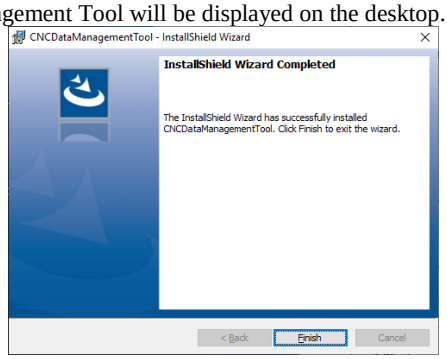


Fig. 2.2.1(b) Installation completion screen

Settings after installation

<2>

When "CNC Data Management Tool" is installed, the following folders are created and the writing results are output to the log file.

"C:\ProgramData\FANUC\CNCDataManagementTool\Log"

By default, this folder cannot be referenced, so make the following settings.

For Windows 7

1. Display the Control Panel from the Start Menu. Select Appearance and Customization from the Control Panel, and then select Show hidden files and folders from the displayed items.

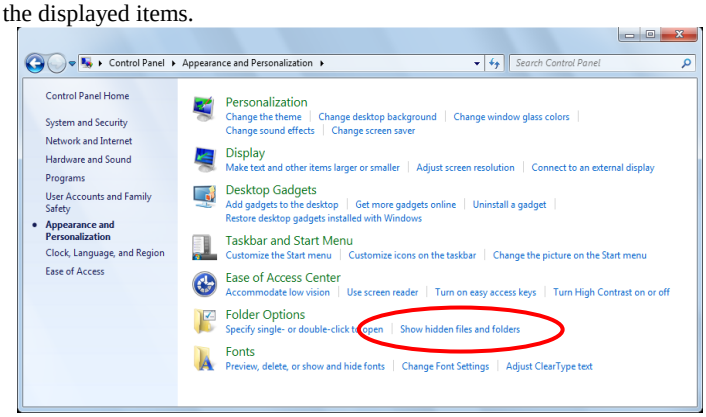


Fig. 2.2.1(c) Appearance and Personalization screen

2. Set the following in Hidden files and folders in Folder Options.

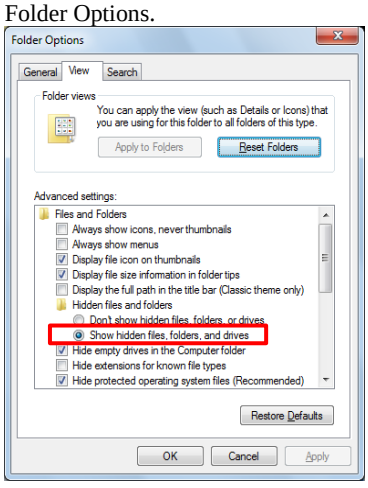


Fig. 2.2.1(d) Option screen

<2>

For Windows 10

Check "Hidden items" on the View tab of Explorer.

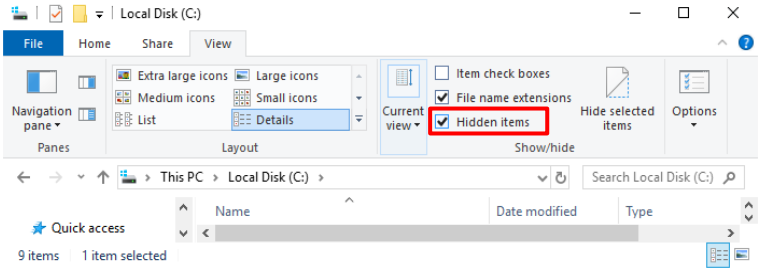


Fig. 2.2.1(e) Explorer

2.2.2 UNINSTALL

This section describes how to uninstall the “CNC Data Management Tool”.

(1) Click the Start button of Windows.
(2) Click Control Panel.
(3) Click Add/Remove Program or Programs and Features. The uninstall screen (Fig. 2.2.2 (a)) is displayed.
(4) Click CNC Data Management Tool.
(5) Click Change/Remove or Uninstall.

Then, proceed with the uninstallation according to the message in the dialog.

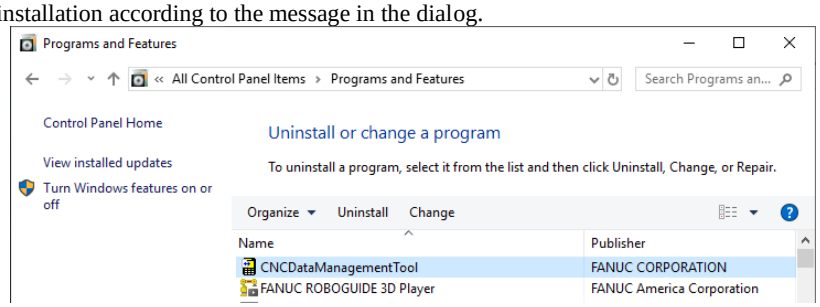


Fig. 2.2.2(a) Uninstall screen

2.2.3 Upgrade

If SETUP.EXE of version 2.2 or later is run on a PC with “CNC Data Management Tool” version 1.3 or later installed, the upgrade will be performed. When upgrading, proceed according to the message (Fig. 2.2.3 (a)). The upgrade overwrites the already installed files and deletes older versions of the files (such as older manuals).

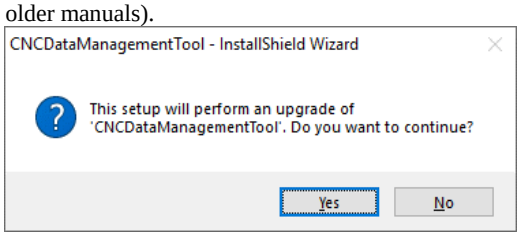


Fig. 2.2.3(a) Upgrade confirmation screen

NOTE
Impossible to upgrade from “CNC Data Management Tool” version 1.2 or earlier. Please uninstall and then reinstall.

2.2.4 Downgrade

<7>

"CNC Data Management Tool" cannot be downgraded. If an older version of SETUP.EXE is run than the installed "CNC Data Management Tool", the following message (Fig. 2.2.4 (a)) will be displayed. If downgrade is necessary, please uninstall and then reinstall.

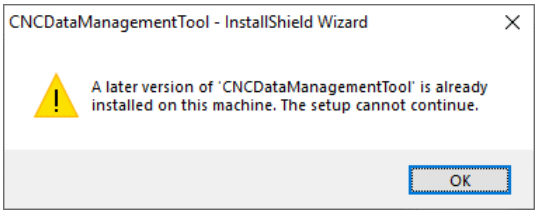


Fig. 2.2.4(a) Downgrade interruption screen

<7>

2.3 SETTINGS FOR COMMUNICATION

2.3.1 Settings for Tool Side

The following values are set as standard values. If the change is necessary, change the setting in the toolbar "Setting" → "Communication setting".

-IP address : 192.168.1.1
-TCP port number : 8193
-Time out : 2

2.3.2 Settings for CNC Side

- PCMCIA settings

This tool assumes that the PCMCIA-LAN card is inserted in the PCMCIA card slot on the front of the CNC and connected to the Ethernet. Therefore, when using PCMCIA to write a file to CNC, no configuration is required.
However, when changing the PCMCIA settings, change the IP address etc. according to the following procedure.

1. Press the function key [SYSTEM].
2. Soft key [PCMCIA LAN] appear. (When there is no soft key, press the continue key.)
3. Press soft key [PCMCIA LAN] to display the setting screen of the "COMMON:SETTING[PCMCIA]".
4. If the available device is "EMBEDDED", press the [OPRT] soft key and then [EMB/PCMCIA] to change the available device to "PCMCIA".

- EMBEDDED settings

When writing a file to the CNC using the built-in port, change the IP address etc. according to the following procedure.

1. Press the function key [SYSTEM].
2. Soft key [EMBED PORT] appear. (When there is no soft keys, press the continue key.)
3. Press soft key [EMBED PORT] to display the setting screen of the "COMMON: SETTING[EMBEDDED]".
4. If the available device is "PCMCIA", press the [OPRT] soft key and then [EMB/PCMCIA] to change the available device to "EMBEDDED".

3 OPERATION OF CNC DATA MANAGEMENT TOOL

3.1 START

By executing the created shortcut or "CNC Data Management Tool" on the start menu, this tool will start and the following screen will be displayed. Switch between write and read by tab operation.

- [PC -> CNC] tab (Write to CNC)

The following startup screen (Fig. 3.1 (a)) is displayed on the [PC -> CNC] tab.
Refer to the "WRITE TO CNC" section for details.

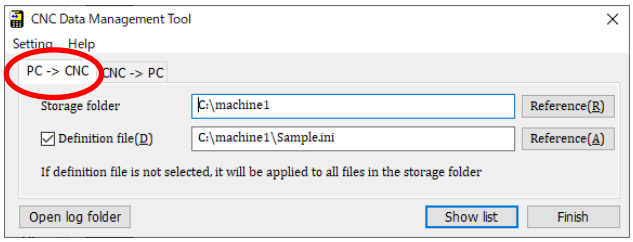


Fig. 3.1(a) Startup screen ([PC -> CNC])

- [CNC -> PC] tab (Read from CNC)

The following startup screen (Fig. 3.1 (b)) is displayed on the [CNC -> PC] tab.
Refer to the "READ FROM CNC" section for details.

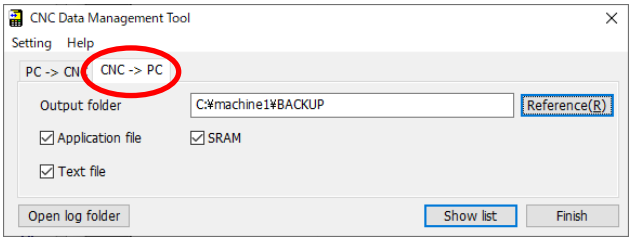


Fig. 3.1(b) Startup screen ([CNC -> PC])

3.2 WRITE TO CNC ([PC -> CNC] TAB)

Write various files to CNC. Do the following before writing starts:

(1) Create a folder to store files for writing to CNC (*1)
(2) Create a file that defines what to write to CNC (*2)
(3) Storage various files to be written to CNC in the folder created in (1)

For convenience of explanation, *1 is sometimes called "storage folder" and *2 is sometimes called “definition file".
Refer to the "Target file" section for "storage folder" and "definition file".
Example) Prepare various files to be written to machine 1

(1) Create a storage folder named "machine1".
(2) Create a "definition file" named "machine1.ini" and save it in the “storage folder”.
(3) Collect the files to be written to CNC in the “storage folder”.

```
C:\
├── \machine1\                 …Storage folder
│     ├── machine1.ini         …Definition file
│     ├── M1_IOCONF.000        …Below, various files
│     ├── M1_PLC1.000
│     ├── M1_CNC-PARA.TXT
│     :
```

When ready, start the tool and select the [PC -> CNC] tab. The following startup screen (Fig.3.2 (a)) is displayed. <5> <6>

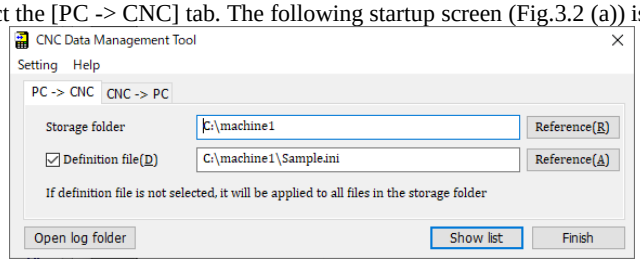


Fig. 3.2(a) Startup screen ([PC -> CNC]) <6>

- Select storage folder

Select the folder where the files to be written to CNC by key input or the [Reference] button. If "Definition file" is not selected, the following will occur depending on the existence of “definition file” in the selected “storage folder”. <6>

(1) If the “Definition file” does not exist

A text file with a default file name and an application file are written to the CNC.

For more information on default file names, see "DEFAULT FILE NAME".

(2) If the “definition file” exist

The file is written to CNC according to the “definition file”.

- Select definition file

Select the “definition file” that specifies the file to be written to CNC by key input or the [Reference] button. The file is written to CNC according to the selected definition file. For details on the definition file, refer to the "DEFINITION FILE" section. <6>

- [Reference] button

A folder or file selection screen will be displayed. Select the desired “storage folder” or “definition file”.

- [Open log folder] button <4>

Open the folder where the log is output. For details on log files, refer to the "LOG OUTPUT" section. <4>

- [Show list] / [Execution] button (The display changes depending on the tool options settings.) <14>

Check the storage folder (or definition file when used) and operate as follows depending on the button state. <14>

[Show list] button : Displays the files to be written in a write screen (Figure 3.2.1(a)). <14>

[Execution] button : Starts writing the files to be written. <14>

- [Finish], [×] button

This tool will end.

3.2.1 Write screen <5>

When clicking the [Show list] button on the startup screen (Fig. 3.2(a)), the following writing screen (Fig. 3.2.1(a)) is displayed. Files for which the check boxes at the left end are turned ON will be written. Files that are turned OFF will not be written to the CNC. The check boxes are all ON in the initial state.

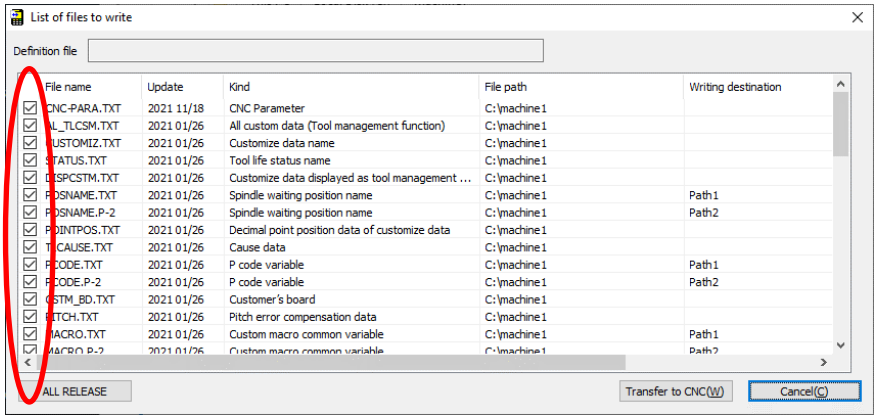


Fig. 3.2.1(a) Write screen <5>

- [ALL RELEASE] button

All check boxes at the left end are turned OFF. Also, the button changes from [ALL RELEASE] to [ALL SELECT].

- [ALL SELECT] button

All check boxes at the left end are turned ON. Also, the button changes from [ALL SELECT] to [ALL RELEASE].

- [Transfer for CNC] button

Writing to the CNC is performed. During writing, a dialog (Fig.3.2.1 (b)) will be displayed to show the progress.

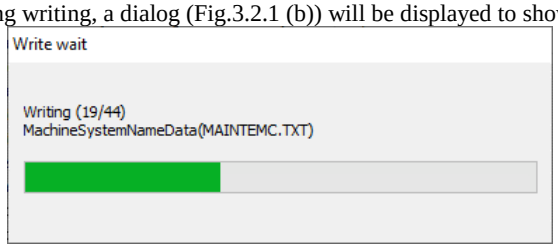


Fig. 3.2.1(b) Write Wait

If all writing is successful, the following message (Fig 3.2.1(c)) will be displayed. Restart the CNC according to the message.

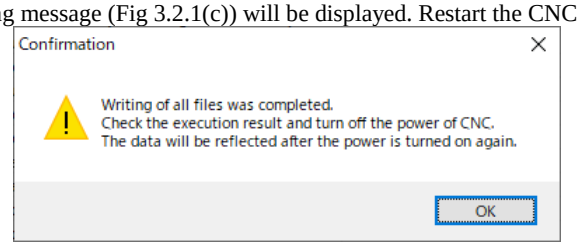


Fig. 3.2.1(c) Writing completed

If any of the files could not be written successfully, the following message (Fig 3.2.1(d)) will be displayed after writing all the files.

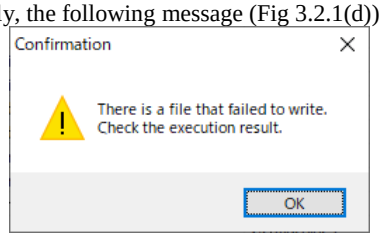


Fig. 3.2.1(d) Write failure

- [Cancel], [×] button

Return to the startup screen (Fig. 3.1 (a)).

- Writing after turning on the power again

Some files, such as the work coordinate system file (EXT_WKZ.TXT), cannot be written normally unless the parameters are confirmed. Therefore, parameter writing is given priority. However, PW0000 "POWER MUST BE OFF" occurs while writing the parameters. When PW0000 occurs, there is a file that cannot be written, so a dialog requesting a CNC restart is displayed (Fig 3.2.1(e)). In that case, restart the CNC.

After the CNC is restarted, the remaining files will be written automatically.

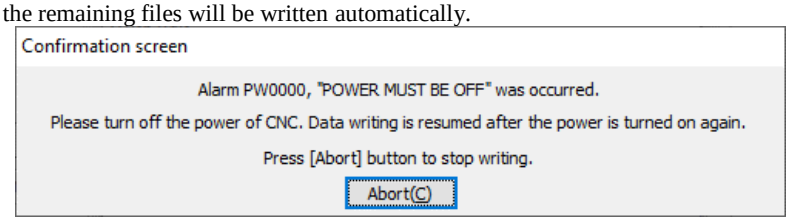


Fig. 3.2.1(e) Screen prompting to restart the CNC

- Writing after turning on the power again (When changing PMC parameter settings)

When writing CNC parameters, if the number of PMC paths or memory type is changed, the PMC alarm "ER09 PMC LABEL CHECK ERROR" may occur after the CNC is restarted. When this alarm occurs, PMC parameters cannot be written. Therefore, if the number of PMC paths or memory type is changed when writing CNC parameters, a dialog prompting to clear the PMC alarm ER09 is displayed (Fig 3.2.1 (f, g)). If this dialog is displayed, according to the message in the dialog and clear the PMC memory. After the CNC is restarted, the remaining files will be written automatically.

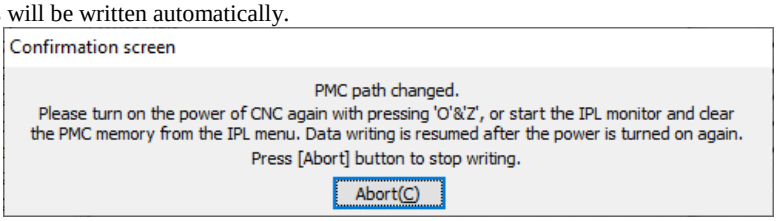


Fig. 3.2.1(f) Screen prompting to restart the CNC (When the number of PMC paths is changed)

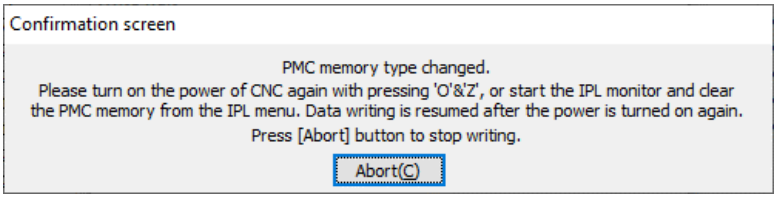


Fig. 3.2.1(g) Screen prompting to restart the CNC (When the memory type of PMC is changed)

- [Abort] button

Writing is interrupted. Return to the startup screen (Fig. 3.2 (a)).

- Result of write execution

The write execution result of the files to be written is displayed on the write screen (Fig.3.2.1 (h)). See success or failure in the execution result column.

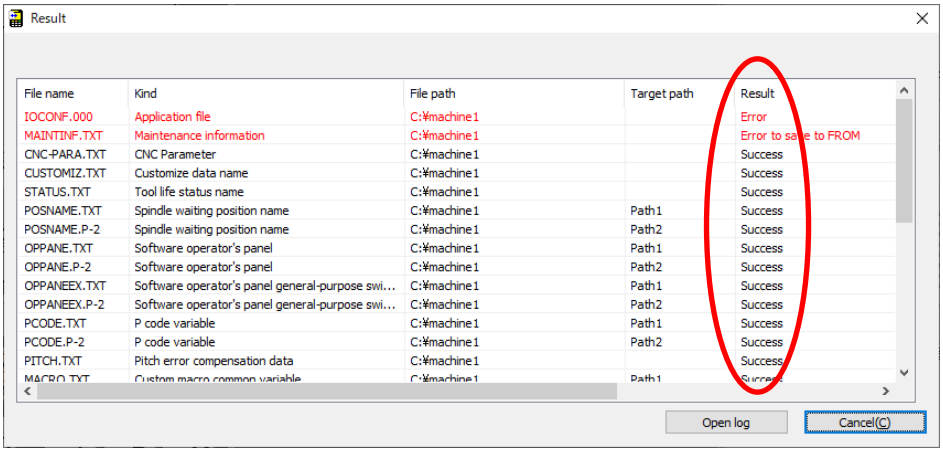


Fig.3.2.1 (h) Write execution result screen

The displayed results and display order are as follows.

Display order	Result	Content	Remarks
1	Error	Write error.	Displayed in red.
2	Error to save to FROM	Error to save maintenance information to FROM.	Displayed in red.
3	Stop	When writing a program file, writing of the program file was interrupted because the same program number was already registered.	
4	Skip	When writing a program file, writing of that program number was skipped because the same program number was already registered.	
5	Success	Write success.	

3.3 READ FROM CNC ([CNC -> PC] TAB)

Read various files from CNC. Do the following before reading starts:

(1) Create a folder to read various files from CNC (*1)

For convenience of explanation, *1 is sometimes called "output folder".

Example) Prepare an output folder for reading various files from machine 1

(1) Create an output folder "BACKUP" as a subfolder of "machine1".

C:\

├── \machine1\BACKUP ⋯Output folder

When ready, start the tool and select the [CNC -> PC] tab. The following startup screen (Fig.3.3 (a)) is displayed.

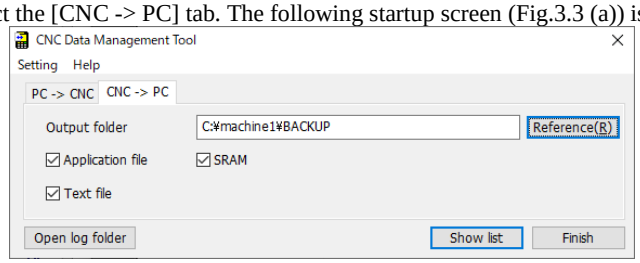


Fig. 3.3(a) Startup screen ([CNC -> PC])

- Select output folder

Select the “output folder" to store various files read from CNC by key input or the [Reference] button. All output files will have default file names. Refer to the "DEFAULT FILE NAME" section for details on default file names.

- Application file, text file, SRAM data

Select the type of file to read from CNC. The target application files and text files differ depending on the model and CNC options. Refer to the "CORRESPONDENCE BY MODEL" section for the files that can be read.

- [Reference] button

A folder selection screen will be displayed. Select the desired “output folder”.

- [Open log folder] button

Open the folder where the log is output. For details on log files, refer to the "LOG OUTPUT" section.

- [Show list] / [Execution] button (The display changes depending on the tool options settings.)

It is enabled by selecting an application file or text file or SRAM data. Check the readable files and operate as follows depending on the button state.

[Show list] button : The readable files of the selected file type are displayed on the read screen (Figure 3.3.1(b)).

[Execution] button : Starts reading the readable files of the selected file type.

3.3.1 Read screen

When click the [Show list] button on the startup screen (Fig.3.3 (a)), the following dialog (Fig.3.3.1 (a)) will be displayed. Get the readable target file from CNC (actual file read is not done).

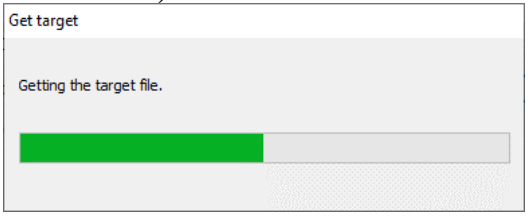


Fig. 3.3.1(a) Getting the target file

CAUTION
It may take some time depending on the CNC options and the number of paths.

When the acquisition of the target file is completed, the following reading screen (Fig.3.3.1 (b)) will be displayed. Files with the leftmost check box checked will be read. Files that are turned off will not read from the CNC. The check boxes are all ON in the initial state.

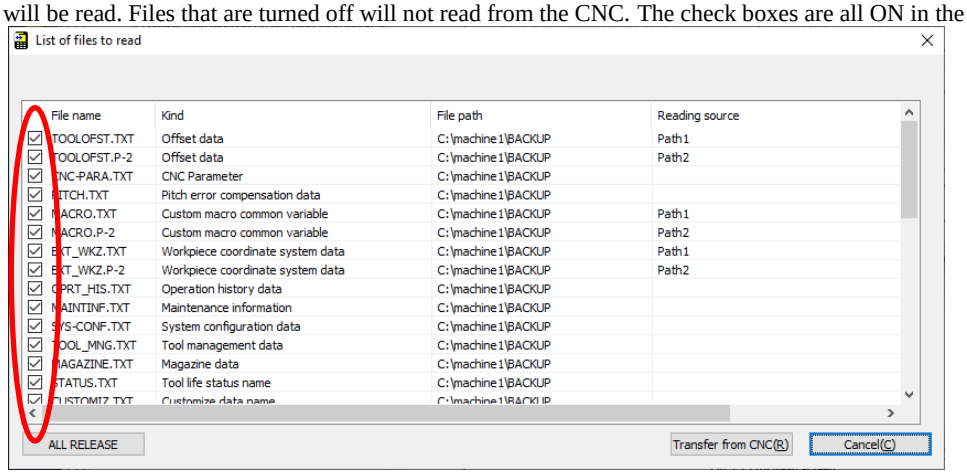


Fig. 3.3.1(b) Read screen

- [ALL RELEASE] button

All check boxes at the left end are turned OFF. Also, the button changes from [ALL RELEASE] to [ALL SELECT].

- [ALL SELECT] button

All check boxes at the left end are turned ON. Also, the button changes from [ALL SELECT] to [ALL RELEASE].

- [Transfer from CNC] button

Reading from the CNC is performed. During reading, a dialog (Fig.3.3.1 (c)) will be displayed to show the progress.

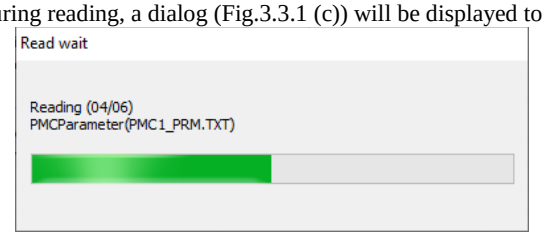


Fig. 3.3.1(c) Read Wait

If all reading is successful, the following message (Fig 3.3.1(d)) will be displayed.

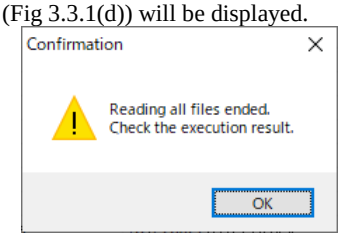


Fig. 3.3.1(d) Read completed

If any one of the files could not be read successfully, the following message (Fig 3.3.1(e)) will be displayed after reading all the files.

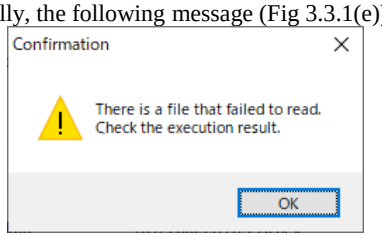


Fig. 3.3.1(e) Read failure

- Result of read execution

The execution result of reading the files to be read is displayed on the reading screen (Fig. 3.3.1 (b)). See success, failure, and skip in the execution result column.

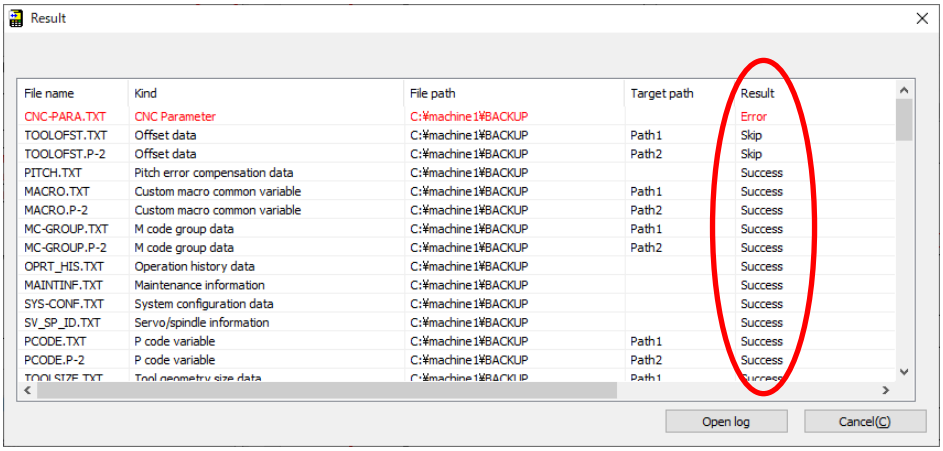


Fig. 3.3.1 (f) Read execution result screen

The displayed results and display order are as follows.

Display order	Result	Content	Remarks
1	Error	Read error.	Displayed in red.
2	Skip	When reading a file, reading of the file was skipped because a file with the same name exists in the output folder.	
3	Success	Read success.	

3.4 SETTING

Change the display language etc. in the toolbar settings.

3.4.1 Language

Select from "Japanese" or "English".

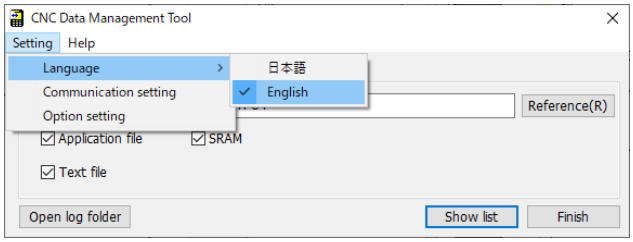


Fig. 3.4.1(a) Language setting

3.4.2 Communication setting

Settings related to the communication can be done.

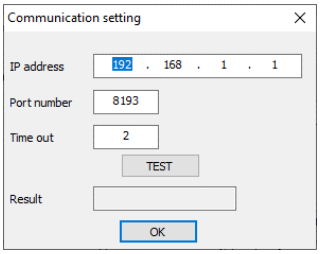


Fig. 3.4.2(a) Communication setting

CNC has the security function that restricts access from Ethernet-connected applications, including this tool. With this function, the access code is set in NC parameter No.10344 (*1). If this access code is set, the tool side must also specify the same access code. Set the access code that is displayed when you press "Ctrl" + "Alt" + "A" on the keyboard at the same time on the communication setting screen. <6>

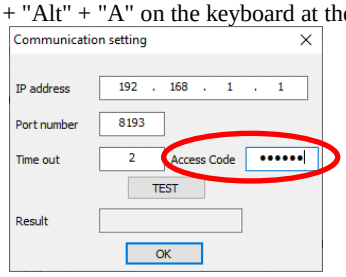


Fig. 3.4.2(b) Access code setting <6>

(*1) Set access code to 0 in this tool for the model which CNC parameters No.10344 is not available for (FS0*i*-D/PD/Mate-D, FS30*i*/31*i*/32*i*-A, FS16*i*/18*i*/21*i*-B). This tool cannot connect with CNC if a value other than 0 is set. <8> <8>

3.4.3 Option setting <7>

Settings related to various settings can be done. (1) to (13) are explained below. <12>

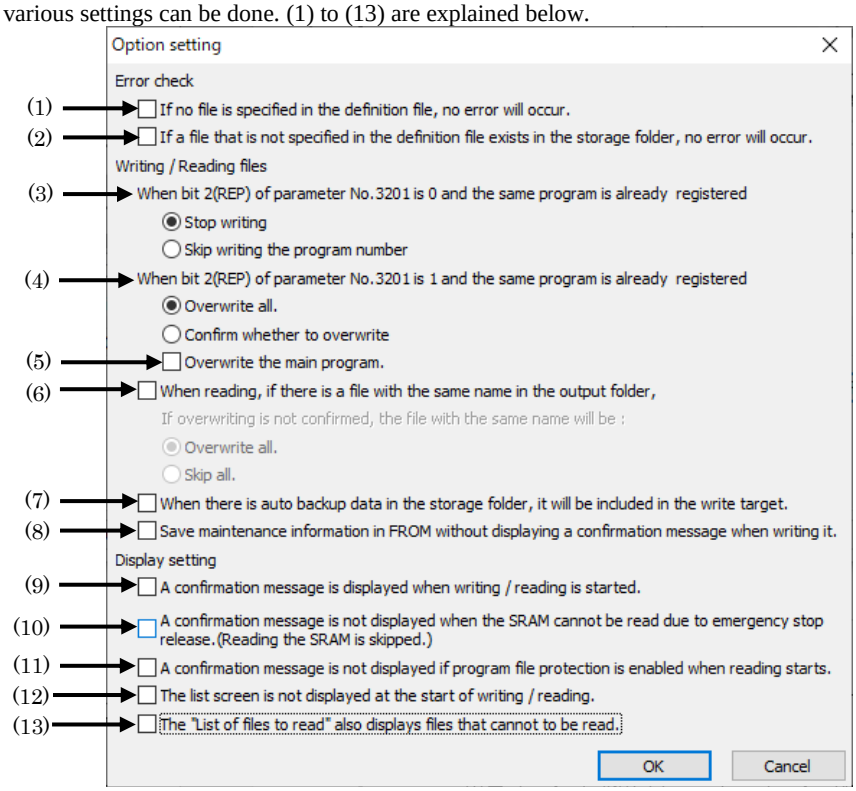


Fig. 3.4.3 (a) Option setting <12>

- Error check <5>

(1) If the specified file does not exist in the definition file, specify can be done whether to make an error or not.
ON : No error occurs
OFF : An error occurs

(2) If a file that is not specified in the definition file exists in the storage folder, specify whether to make an error or not.
ON : No error occurs
OFF : An error occurs <5>

- Writing / Reading files

(3) In writing a program file, when bit 2(REP) of parameter No.3201 is 0 and the same program is already registered, select can be done whether to interrupt or skip writing. When the writing of the registered program is stopped or skipped and the writing of all files is completed, the following message (Fig. 3.4.3 (b), Fig. 3.4.3 (c)) is displayed.



Fig. 3.4.3(b) stop message

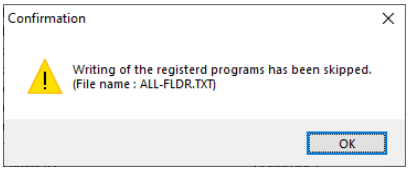


Fig. 3.4.3(c) skip message

(4) In writing a program file, when bit 2(REP) of parameter No.3201 is 1 and the same program is already registered, select whether to overwrite unconditionally or confirm overwriting each time can be done. If select “Confirm whether to overwrite”, the following message (Fig. 3.4.3(d)) will be displayed.

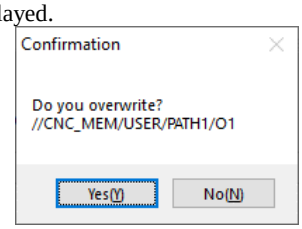


Fig. 3.4.3(d) Confirmation of overwrite

(5) In writing a program file, when bit 2(REP) of parameter No.3201 is 1, the main program can be overwritten.

ON : Overwrite the main program. When writing starts, the following message (Fig. 3.4.3(e)) will be displayed. Change the CNC mode according to the message.

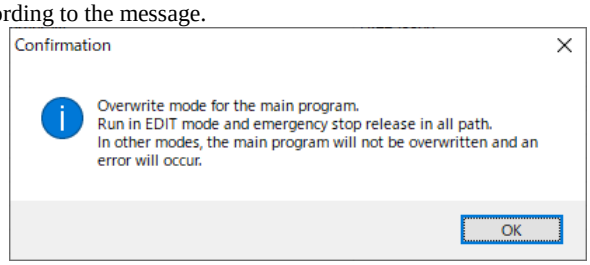


Fig.3.4.3 (e) Mode confirmation

OFF: The main program will not be overwritten. When the program file contains the main program, an error occurs when writing.

NOTE

1 In the above "when the same program is already registered" behavior, this tool checks for the existence of registered programs. The following cases are determined to be program by this check.
- Address "O"
- Program name following ‘<’ at the beginning of the block
(The character string following ‘<’ in the middle of the block is not determined as the program name)

If it cannot be determined in the check process, a tool occurs an error (error code: 3003). For the log when this error occurs, refer to the "LOG OUTPUT" section, and for the corrective action, refer to the "ERROR CODE LIST" section.

2 If the target CNC is FS16i/18i/21i-B, the registered programs will not be checked. In FS16i/18i/21i-B, when the registered program is written, it will be as follows depending on the setting of bit 2(REP) of parameter No.3201.
1: Overwritten
0: Error (error code: 1005)

NOTE

3 To enable overwriting of the main program, in addition to the option settings of CNC Data Management Tool, all of the following (i) to (iv) must be satisfied when writing the program file. Otherwise, an error (error code: 1005, detail code: 5) will occur when writing the main program.
   (i) Emergency stop cancellation for all paths
   (ii) Select EDIT mode for all paths
   (iii) Bit 2(REP) of parameter No.3201 is 1
   (iv) Program can be created
   CNC Data Management Tool first creates an empty program in the CNC, and writes the program after making it as a temporary main program. Therefore, the number of registrable programs in CNC must be 1 or more, and the program capacity must be 1 KB or more. An empty program is created with the name "CNC_MNG_TOOL_TMP" under the CNC program folder //CNC_MEM/USER/. Do not use this program because it is a reserved word for CNC Data Management Tool. Also, "CNC_MNG_TOOL_TMP" will be deleted after program writing is completed.
4 FS0i-D/PD/Mate-D, FS16i/18i/21i-B cannot overwrite the main program.

<11>

(6) In reading a file, if a file with the same name already exists in the output folder, specify can be done whether to confirm overwriting.
   ON : Do not confirm overwrite. However, choose to overwrite or skip can be done.
   OFF : The following message (Fig.3.4.3 (f)) will be displayed.

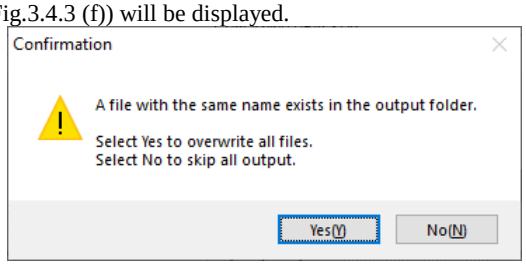


Fig.3.4.3 (f) Confirmation of overwrite

(7) When there is auto backup data in the storage folder, specify whether it will be included in the write target.
   ON : Target auto backup data for writing.
   OFF : Do not write auto backup data.
However, if auto backup data is specified in the definition file, it will be written regardless of this setting.

<7>

NOTE

1 Auto backup data can only be written by the CNC from which the data was retrieved. If write to another CNC, when restore with the boot system, "SYSTEM LABEL ERROR" will occur and the CNC will not be able to start.
2 This tool can write only auto backup data read by this tool. Files output by the boot system or other functions cannot be written.
3 Files read with version 2.1 or earlier of this tool cannot be written. Also, it cannot be written with this tool. Writing will result in an error (error code: 3002).
4 Do not write the auto backup data read by this tool except this tool.(Boot system, etc.)

<7>

(8) When writing maintenance information (MAINTINF.TXT), it is also possible to specify whether to display a message to confirm saving of maintenance information.
   ON : Do not display message. Maintenance information is automatically saved.
   OFF: The following message (Fig. 3.4.3(g)) will be displayed.

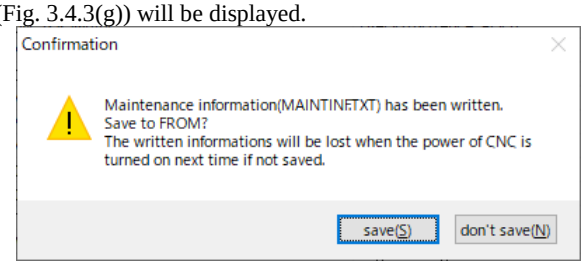


Fig. 3.4.3(g) Confirmation screen for saving maintenance information

<11>

Depending on the CNC model and software version, the CNC Data Management Tool may not be able to save maintenance information to FROM. In this case, the option setting in (8) becomes invalid and the following message (Fig.3.4.3(h)) is displayed. Follow the message and save the contents on the maintenance information screen. <11>

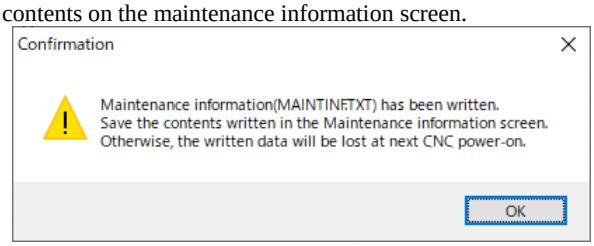


Fig.3.4.3 (h) Screen prompting to save maintenance information <11>

- Display setting

(9) When writing / reading starts, select can be done whether to display the confirmation message dialog.

ON : The following message (Fig.3.4.3 (i)) will be displayed.

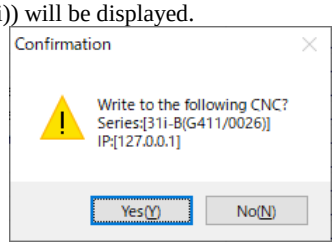


Fig.3.4.3 (i) Confirmation message (In case of writing)

OFF : Do not display message

(10) Set whether to confirm retries or skip reading when reading SRAM data in the emergency stop release state. This setting is valid for models that do not allow SRAM data reading in the emergency stop release state. Refer to the "CORRESPONDENCE BY MODEL" section to see whether SRAM data can be read when the emergency stop is released. <6>

ON : Does not confirm retry. SRAM data reading is skipped.

OFF: The following message (Fig.3.4.3 (j)) is displayed.

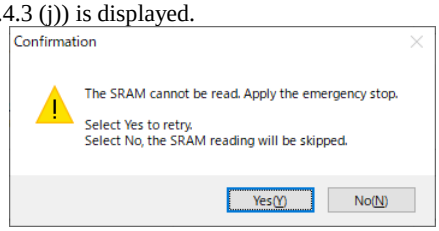


Fig.3.4.3 (j) Confirmation of retry <6>

(11) Set whether or not to display a confirmation message when the CNC program protection parameter is valid in reading a program. <10>

ON : Do not display confirmation message.

OFF: The following message (Fig 3.4.3(k)) is displayed depending on the program protection parameter settings.

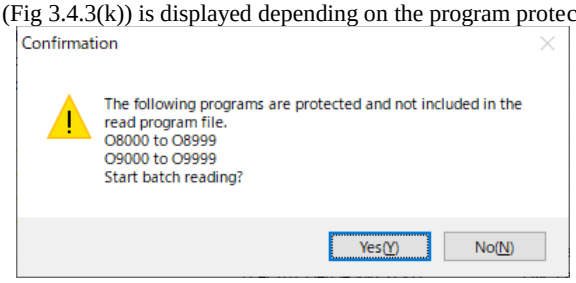


Fig.3.4.3(k) Confirmation of program protection <10>

(12) When writing / reading starts, select can be done whether to display the list screen (Write screen: (Fig.3.2.1(a)), Read screen: (Fig.3.3.1(b))). However, in either case, when the write / read is completed, the execution result screen (Fig.3.2.1 (i), (Fig.3.3.1 (f)) is displayed. <5> <8> <8>

ON : The list screen is not displayed. Click the [Execution] button on the startup screen to start writing/reading. <14>

OFF: Click the [Show list] button on the startup screen to display the list screen. <14>

(*)When turned ON, the [Execution] button on the startup screen becomes the [Execution] button. <14>

(13) Set whether to display unreadable files on the list of files to read.

Unreadable files are displayed as shown below.

- Checkbox hidden
- Background color gray

ON : The list of files to read also displays unreadable files.

OFF : The list of files to read does not display unreadable files.

Example)

If the workpiece coordinate system data cannot be read from the connected CNC, the following list will be displayed.

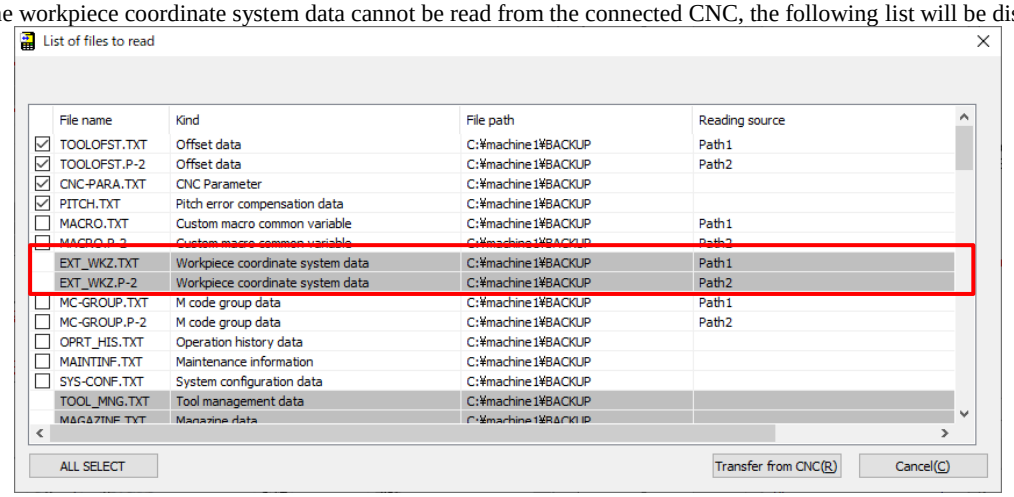


Fig.3.4.3(l) Example of displaying unreadable files on the list of files to read

On the result screen that is displayed after reading, unselected files and excluded unreadable files are displayed as shown below.

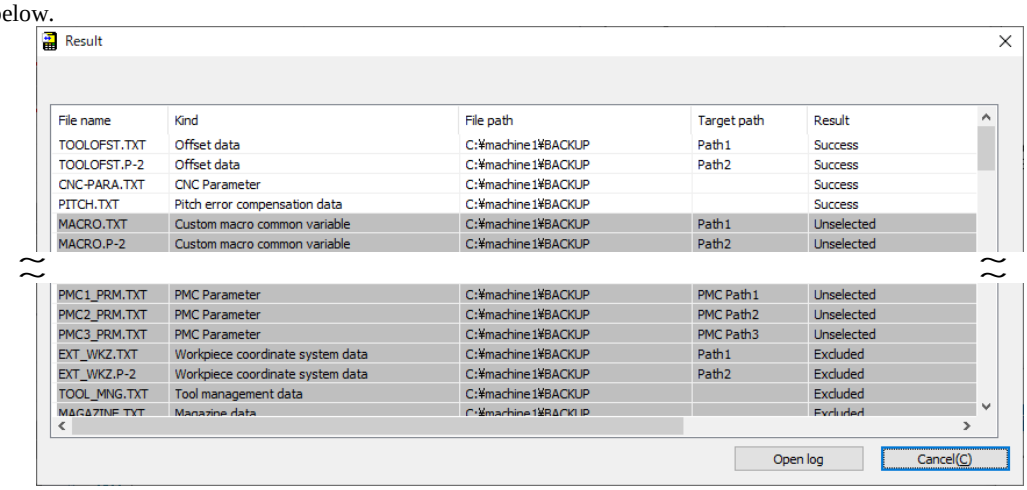


Fig.3.4.3(m) Example of displaying unselected files and excluded files on the reading result screen

When displaying a list of unreadable files, the displayed results and display order are as follows.

Display order	Result	Content	Remarks
1	Error	Read error.	Displayed in red.
2	Skip	When reading a file, reading of the file was skipped because a file with the same name exists in the output folder.	
3	Success	Read success.	
4	Unselected	The checkbox on the list of files to read is unchecked.	Displayed with a gray background.
5	Excluded	This file cannot be read by CNC Data Management Tools. Possible reasons are: - The connected CNC has no functions or options. - Depending on the model and CNC software version, the CNC may not support output.	

3.5 Help

(1) Version

The version is displayed.

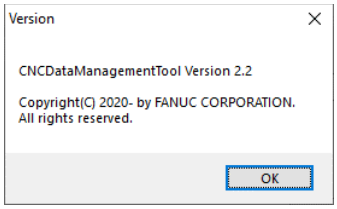
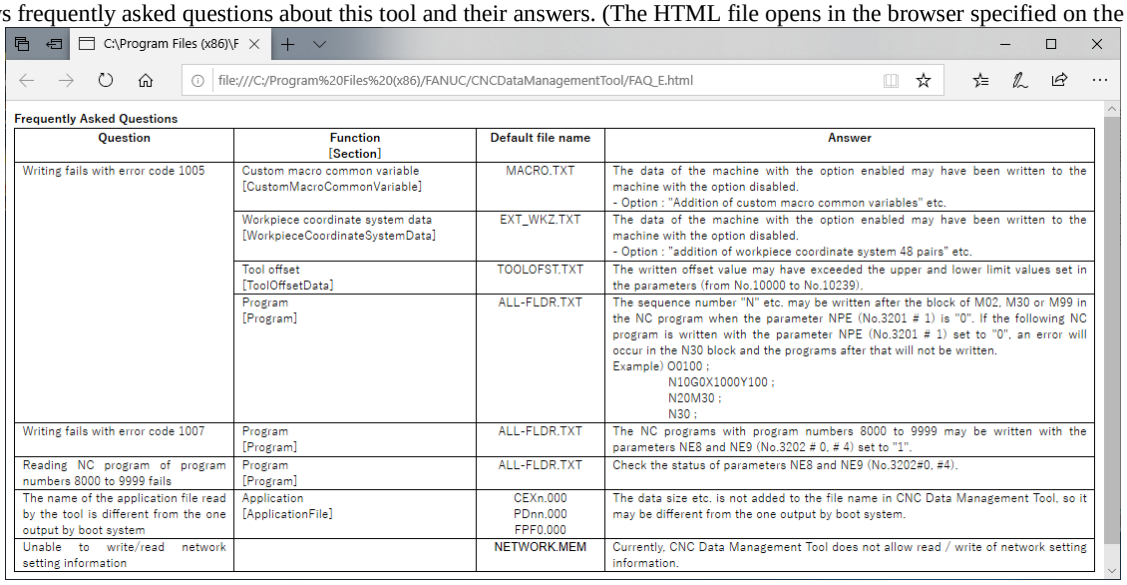


Fig. 3.5(a) Version

(2)FAQ

Shows frequently asked questions about this tool and their answers. (The HTML file opens in the browser specified on the PC)



Frequently Asked Questions

Question	Function [Section]	Default file name	Answer
Writing fails with error code 1005	Custom macro common variable [CustomMacroCommonVariable]	MACRO.TXT	The data of the machine with the option enabled may have been written to the machine with the option disabled. - Option : "Addition of custom macro common variables" etc.
	Workpiece coordinate system data [WorkpieceCoordinateSystemData]	EXT_WKZ.TXT	The data of the machine with the option enabled may have been written to the machine with the option disabled. - Option : "addition of workpiece coordinate system 48 pairs" etc.
	Tool offset [ToolOffsetData]	TOOLOFST.TXT	The written offset value may have exceeded the upper and lower limit values set in the parameters (from No.10000 to No.10239).
	Program [Program]	ALL-FLDR.TXT	The sequence number "N" etc. may be written after the block of M02, M30 or M99 in the NC program when the parameter NPE (No.3201 # 1) is "0". If the following NC program is written with the parameter NPE (No.3201 # 1) set to "0", an error will occur in the N30 block and the programs after that will not be written. Example) O0100 ; N10G0X1000Y100 ; N20M30 ; N30 ;
Writing fails with error code 1007	Program [Program]	ALL-FLDR.TXT	The NC programs with program numbers 8000 to 9999 may be written with the parameters NE8 and NE9 (No.3202 # 0, # 4) set to "1".
Reading NC program of program numbers 8000 to 9999 fails	Program [Program]	ALL-FLDR.TXT	Check the status of parameters NE8 and NE9 (No.3202#0, #4).
The name of the application file read by the tool is different from the one output by boot system	Application [ApplicationFile]	CEXn.000 PDnn.000 FPF0.000	The data size etc. is not added to the file name in CNC Data Management Tool, so it may be different from the one output by boot system.
Unable to write/read network setting information		NETWORK.MEM	Currently, CNC Data Management Tool does not allow read / write of network setting information.

Fig. 3.5(b) FAQ file (for Microsoft Edge)

4 TARGET FILE <5>

4.1 STORAGE FOLDER

Store the file to be written to the CNC in the “storage folder”. The files stored in this folder will be written to the CNC.

NOTE
If the "storage folder” is selected and the "definition file" is not selected in the startup screen (Fig. 3.1 (a)), and the “definition file” does not exist in the “storage folder”, the default file name and application file are written to the CNC. Do not store files that do not need to write in the “storage folder”.

4.2 DEFAULT FILE NAME

<5>

The default file name is handled as follows when writing and reading.

Writing

If the definition file does not exist, the file with the default file name will be written to the CNC. This applies only to text files. The application file is automatically determined and written based on the file information written in the file, regardless of the file name.

<5>

Reading

The application files, text files, and SRAM data read from the CNC are saved as the default file names.

<6>

Default file name (Text file)

The default file name of the text files are as follows.

<5>

Table 4.2 (a) List of Default file name (other than FS16*i*/18*i*/21*i*-B)

No.	Default file name	Data kind	Path	Remarks
1	ALL-PROG.TXT	Program	✓	(*1) <5>
-	ALL-FLDR.TXT	Program		(*1) <5>
2	TOOLOFST.TXT	Tool offset	✓	
3	CNC-PARA.TXT	Parameter		
4	PITCH.TXT	Pitch error compensation data		
5	COMP3D.TXT	Three-dimensional error compensation data		(*2) <5>
6	CMP3DROT.TXT	Three-dimensional rotary error compensation data		<2>
7	MACRO.TXT	Custom macro common variable	✓	
8	EXT_WKZ.TXT	Workpiece coordinate system data	✓	
9	TOOL_MNG.TXT	Tool management data		
10	MAGAZINE.TXT	Magazine data		
11	STATUS.TXT	Tool life status name		
12	CUSTOMIZ.TXT	Customize data name		
13	DISPCSTM.TXT	Customize data displayed as tool management data		
14	POSNAME.TXT	Spindle waiting position name	✓	
15	POINTPOS.TXT	Decimal point position data of customize data		
16	TOOLGEOM.TXT	Tool geometry data		
17	WSEC_VAL.TXT	Workpiece setting error value	✓	
18	TOOLLIFE.TXT	Tool life management data	✓	
19	MAINTINF.TXT	Maintenance information		
20	MAINTENA.TXT	Periodic maintenance data		(*2) <5>
21	SEC_GEOM.TXT	Second geometry tool offset	✓	
22	FIXTURE.TXT	Rotary table dynamic fixture offset		

No.	Default file name	Data kind	Path	Remarks	
23	RTMMACRO.TXT	Real time custom macro variable	✓	(*2)	<5>
24	DIDOENBL.TXT	PMC signal protect	✓	(*2)	<5>
25	TLCAUSE.TXT	Cause data			
26	MAINTEMC.TXT	Machine system name data		(*2)	<5>
27	PCODE.TXT	P code variable	✓	(*8)	<11>
28	MC-GROUP.TXT	M code group data	✓		
29	TOOLSIZE.TXT	Tool geometry size data	✓		
30	CSTM_BD.TXT	Customer’s board			
31	OHIS_SIG.TXT	Operation history signal section data			
32	3D-INTER.TXT	3D interference check data			
33	AL_TLMGR.TXT	All tool data	✓		
34	AL_TLCSM.TXT	All custom data			
35	MACHCON.TXT	Machine configuration	✓		
36	MCS_SET.TXT	Machining condition data	✓		<9>
37	SCROLL-M.TXT	Scroll waiting M-code			<9>
38	SAC-PARA.TXT	Table data of smart spindle-load control			<9>
39	WORKSHFT.TXT	Workpiece coordinate system shift	✓	(*4)	<10>
40	OPPANE.TXT	Software operator's panel	✓	(*5), (*7)	<10>
41	OPPANEEX.TXT	Software operator's panel general-purpose switch	✓	(*6), (*7)	<10>
42	TOOLDATA.TXT	Tool data	✓		<4>
43	LAS_CON.TXT	Cutting condition data	✓		<4>
44	PMCn_PRM.TXT	PMC parameter		n: path number	
45	PMCS_PRM.TXT	PMC parameter (Dual Check Safety)			
46	CNCIDNUM.TXT	CNC ID information		Read only	<5>
47	SYS_ALM.TXT	System alarm information		Read only	
48	SV_SP_ID.TXT	Servo/spindle information		Read only	
49	SYS-CONF.TXT	System configuration data		Read only	
50	OPRT_HIS.TXT	Operation history data		Read only	
51	TLATAHIS.TXT	Attachment history data		Read only	
52	TLDTAHIS.TXT	Detachment history data		Read only	
53	MSRHSTRY.TXT	Machine state history data		Read only	
54	PW-OFS.TXT	Power calibration coefficient history	✓	Read only	
55	P-ACT.TXT	Parts run hour	✓	Read only	
56	LSALM-H.TXT	History of the alarm related to discharge		Read only	
57	LALMWNG.TXT	Laser alarm history		Read only	
58	WARMUP.TXT	Warm up history		Read only	<5>
59	CNC-DGN.TXT	Diagnosis data		Read only(*3)	<8>
60	iHMI-VER.TXT	iHMI version information		Read only	<9>
61	PTO-HIST.TXT	Path table operation history function		Read only	
62	OPRT_BAK.TXT	Operation history backed up in FROM		Read only	
63	LMSUR000.CSV	Auto measurement data		Read only	
64	LTRND000.CSV	Laser trend data		Read only	<9>
65	ENB_FUNC.TXT	Enabled function		Read only	<13>

Table 4.2 (b) List of Default file name (FS16*i*/18*i*/21*i*-B)

No.	Default file name	Data kind	Path	Remarks
1	PROGRAM.ALL	Program	✓	(*1)
2	TOOLOFST.DAT	Tool offset	✓	
3	CNCPARAM.DAT	Parameter	✓	
4	PITCHERR.DAT	Pitch error compensation data	✓	
5	MACROVAR.DAT	Custom macro common variable	✓	
6	WORK-G54.DAT	Workpiece coordinate system data	✓	
7	MC-GROUP.DAT	M code group data	✓	
8	OPT-HIST.DAT	Operation history data	✓	
9	WORKSHFT.TXT	Workpiece coordinate system shift	✓	(*4)
10	OPPANE.TXT	Software operator's panel	✓	(*5), (*7)
11	OPPANEEX.TXT	Software operator's panel general-purpose switch	✓	(*6), (*7)
12	PCODE.TXT	P-code variable	✓	(*8)

(*1) For FS0*i*-D/PD/Mate-D, FS16*i*/18*i*/21*i*-B and FS30*i*/31*i*/32*i*-A (STEP2, 3), when reading a file, it is read in the program format and the file name is ALL-FLDR.TXT (PROGRAM.ALL for only FS16*i*/18*i*/21*i*-B). For other models, the file is read in all programs and folder formats, and the file name is ALL-FLDR.TXT.

Only FS30i/31i/32i-A (STEP2, 3), the following is different from other models.

- The file is created for each CNC folder and the file name is ALL-PROG_n.TXT ("n" is a serial number from 1).
- The read CNC folder information is output at the beginning of the file.

Example) When reading from the CNC folder "//CNC_MEM/USER/PATH1/", the file will be as follows.

```
//CNC_MEM/USER/PATH1/%
O0001
N01 G90 G01 X50.0 Y50.0 F1000.
N02 X100.0 Y100.0
N03 M99;
%
```

When writing this file to CNC with "CNC Data Management Tool", write to the folder at the beginning of the file. This specification is also valid for other models. However, depending on the model and file, the operation will be as follows.

- If a folder is specified in the definition file, or if a folder is specified in the ALL-FLDR.TXT (All programs and folder formats), that specification has priority.
- If the nonexistent folder is specified, an error will occur on 0i -MODEL D, 0i Mate-MODEL D and 30*i*/31*i*/32*i*-MODEL A (STEP2, 3). For other models, the specified folder will be created.
- In FS0*i*-D/PD/Mate-D and FS30*i*/31*i*/32*i*-A (STEP2, 3), if "//CNC_MEM/USER/PATH1/" and "//CNC_MEM/USER/PATH2/" are specified, they will be written to path 1 and 2, respectively.

When writing with "F INPUT" on the CNC program edit screen or program list screen, the folder information at the beginning is ignored.

(*2) Read only in emergency stop or edit mode. In other states, it cannot be read and an error occurs.

(*3) Basically, it is read in the same format as the CNC parameter, but it may be P/H/B depending on the data format. The contents of the data to be read are as follows.

Address	Data	Address	Data
N	Diagnosis data number	T	Machine group number
Q	Data type (2 = Diagnosis data)	P	Data (decimal)
A	Servo axis number	H	Data (hexadecimal)
S	Spindle axis number	B	Data (8-bit data)
L	The number of paths.		

Example 1) In the case of systematic 8-bit data

```
N00000Q2L1B00001000L2B00000000
```

Example 2) Servo type decimal data

```
N00319Q2A1P3000A2P3000A3P3000A4P3000A5P3000A6P0A7P0A8P0A9P0A10P0A11P0A12P0
```

Example 3) In the case of spindle type hexadecimal data

```
N01556Q2S1HFFFFFFFFS2H00000000S3H00000000S4H00000000
```

(*4)This is read in the format of programmable data input (G10).

Example)

```
%
G10P0X1.11Y2.22Z3.33
M02
%
```

Be sure to write the setting value with a decimal point. If there is no decimal point, an error (error code: 3005 detail code: 3) will occur.

(*5) This is read in the following format separated by TAB code (0x09).

Example)

NAME	DATA	COMMENT
MODE	0	MDI
HANDLE_AXIS	0	HX
HANDLE_MULT	0	*1
RAPID_OVRD	0	100%
JOG_OVRD	22	100%
FEED_OVRD	5	50%
BLOCK_SKIP	0	OFF
SINGLE_BLOCK	1	ON
MACHINE_LOCK	0	OFF
DRY_RUN	1	ON
PROTECT_KEY	0	PROTECT
FEED_HOLD	0	OFF
PARAMETER		
N07200P01111111		
N07210P0		
:		

NAME, DATA, COMMENT, PARAMETER the meaning of is explained below.

- NAME

Soft switch functions. Each has the following meaning.

NAME	Meaning
MODE	Mode
HANDLE_AXIS	Handle axis
HANDLE_MULT	Handle multiple / Incremental multiple
RAPID_OVRD	Rapid override
JOG_OVRD	Jog override
FEED_OVRD	Feed override
BLOCK_SKIP	Block skip
SINGLE_BLOCK	Single block
MACHINE_LOCK	Machine lock
DRY_RUN	Dry run
PROTECT_KEY	Protect key
FEED_HOLD	Feed hold

- DATA

This is the setting value of each soft switch. See the "COMMENT" section below for the meaning of the values.

- COMMENT

Indicates the meaning of DATA. Ignored on write. For each function the correspondence between DATA and COMMENT is as follows.

MODE	
DATA	COMMENT
0	MDI
1	MEM
2	EDIT
3	HNDL or INC
4	JOG
5	REF

HANDLE_AXIS	
DATA	COMMENT
-1	- (*)
0	HX
1	HY
2	HZ
3	H4

HANDLE_MULT	
DATA	COMMENT
0	*1
1	*10
2	*100

RAPID_OVRD	
DATA	COMMENT
0	100%
1	50%
2	25%
3	F0

(*) Without the manual handle feed function. Comments are not output when reading.

JOG_OVRD									
DATA	COMMENT	DATA	COMMENT	DATA	COMMENT	DATA	COMMENT	DATA	COMMENT
0	0%	5	0.37%	10	2.0%	15	10.0%	20	52.0%
1	0.1%	6	0.52%	11	2.7%	16	14.0%	21	72.0%
2	0.14%	7	0.72%	12	3.7%	17	20.0%	22	100%
3	0.2%	8	1.0%	13	5.2%	18	27.0%	23	140%
4	0.27%	9	1.4%	14	7.2%	19	37.0%	24	200%

FEED_OVRD									
DATA	COMMENT	DATA	COMMENT	DATA	COMMENT	DATA	COMMENT	DATA	COMMENT
0	0%	5	50%	10	100%	15	150%	20	200%
1	10%	6	60%	11	110%	16	160%		
2	20%	7	70%	12	120%	17	170%		
3	30%	8	80%	13	130%	18	180%		
4	40%	9	90%	14	140%	19	190%		

BLOCK_SKIP, SINGLE_BLOCK, MACHINE_LOCK, DRY_RUN, FEED_HOLD	
DATA	COMMENT
0	OFF
1	ON

PROTECT_KEY	
DATA	COMMENT
0	PROTECT
1	RELEASE

- PARAMETER

  Related parameters of the target path are read. The relevant parameters are:

  - Bits 0(OP1) to 6(OP7) of parameter No.7200
  - Parameter Nos.7210 to 7217

Writing is performed in the order of parameters and software operator's panel status. After writing parameters, the software operator's panel status is written only for functions whose OP1 to OP7 status is "1" (enabled). For the correspondence between each function and Bits 0(OP1) to 6(OP7) of parameter No.7200, refer to the parameters in the "Software Operator's Panel" section of the "Connection Manual (Function) ".

(*6) This is read in the following format separated by TAB code (0x09).

Example)

```
NO    NAME                              DATA
1     Name of general-purpose switch 1  1
2     Name of general-purpose switch 2  1
≈                                       ≈
16    Name of general-purpose switch 16 1

PARAMETER
N07201P00000001
N07205P0
:
```

NO, NAME, DATA, PARAMETER The meaning of is explained below.

- NO

  Software operator's panel it is the number of general-purpose switch 1~16.

- NAME

  This is the name of the software operator's panel general-purpose switches 1 to 16 set in the CNC parameters. It is read out in 8 characters (8 bytes). Refer to the code correspondence table in the "Software Operator's Panel" section of the "Connection Manual (Function)" for the code of the read character. When writing the read file, the parameters (names of software operator's panel general-purpose switches 1 to 16 (1st to 8th characters)) are also changed. For details on the parameters to be changed, refer to the parameters in the "Software Operator's Panel" section of the "Connection Manual (Function)". If the setting is to use double-byte characters (CNC parameter JPC (No.7201#0) is 1), double-byte characters will not be displayed on the read text file. However, if the target CNC is set to use double-byte characters when writing, it will be restored to the CNC as double-byte characters.

- DATA

  Represents the data currently set for each soft switch. 1 indicates ON, 0 indicates OFF. In FS16i/18i-B, when function “Software operator’s panel general purpose switch expansion“ is enabled, "DATA" of general-purpose switches 9 to 16 cannot be read or written. Warning message notifying that read or write is not possible is displayed when the read or write is complete.

- PARAMETER

  Related parameters of the target path are read. When writing, parameters are written first, just like the software operator's panel. The relevant parameters are:
  - Bits 0(JPC), 3(CHS), 4(OPP) of parameter No.7201
  - Bit 1(GPS) of parameter No.7201(*)
  - Parameter Nos.7205 to 7208

  (*) FS0i-D/PD/Mate-D only

(*7) Software operator's panel and software operator's panel general-purpose switch files read by this tool can be written to CNC only by this tool. When writing a memo unrelated to the data in the read file, please enter ' (apostrophe) at the beginning of the line. Anything after that is treated as a comment. When there is data not to be written, comment out the same or delete the target line. Lines without a ' (apostrophe) may be written as data by the tool, resulting in an error or incorrect data.

(*8) The format and restrictions for reading and writing P-code variable with this tool are described below.

- P-CODE read/write format

  The read/write format conforms to CNC F output, but the format for FS16i/18i/21i-MODEL B differs from the format for other models in the following points.
  - 16i/18i/21i-MODEL B dedicated format is read after compatible format with other models.
  - Comments in P-code variable data are read by default. (Other models conform to the setting of Bit 0(MCO) of parameter No.6019.)

  Example of P-CODE output file for FS16i/18i/21i-MODEL B

```
%
G10L86P10000(3FF0000000000000)
G10L86P10001(4000000000000000)
:
M02
%                              ; Up to here compatible format with other models

------------------------------ ; Dedicated format from here
PcodeNo,Val,Dec
10000,100000000,8
10001,200000000,8
:
%

------------------------------ ; Comment from here
P10000(1.0)
```

```
P10001(2.0)
:
%
```

The following table shows how to write data in formats compatible with other models and dedicated formats.

Model to write	Compatible format	Dedicated format
FS16i/18i/21i-MODEL B	ignored	written
FS16i/18i/21i-MODEL B 以外	written	ignored

- Restrictions on reading and writing P-code variable
  - In "F OUTPUT" on CNC screen, when bit 2 (MV2) of CNC parameter No.9034 is 1 and a P-code variable is output, the variable names of custom macro common variables #500 to #549 are also output. In addition, the variable names of custom macro common variables #500 to #549 read as P-code variable can be written to the CNC by “F INPUT” on CNC screen. In CNC Data Management Tool, even if bit 2 (MV2) of CNC parameter No.9034 is set to 1, the variable names of custom macro common variables #500 to #549 may not be read/write as P-code variable depending on the connected CNC. When reading and writing the variable names of custom macro common variables #500 to #549 with CNC Data Management Tool, read and write them as custom macro common variables.
  - P-code variable read from FS16i/18i/21i-MODEL B can be written to other models, but files read from models other than FS16i/18i/21i-MODEL B cannot be written to FS16i/18i/21i-MODEL.
  - When writing floating-point format P-code variable read from FS16i/18i/21i-MODEL B to other models, minute values may differ from the values set in FS16i/18i/21i-MODEL B.

NOTE

1 When writing, specify the following extension in the path data file depending on the writing path. When reading, the extension is added in the same way depending on the reading path.

Target path	Extension
1	.TXT
2	.P-2
3	.P-3
~	
10	.P10

Example) Offset data

TOOLOFST.TXT…Written to Path1.

TOOLOFST.P-2…Written to Path2.

When there is a loader control path or peripheral axes, specify the extension as follows according to the target to be written. Similarly, the following extensions are added when reading.

Target	Extension
Loader path 1	.PL1
Loader path 2	.PL2
Peripheral axis control 1	.PA1
Peripheral axis control 2	.PA2
Peripheral axis control 3	.PA3

However, the loader path of FS16i/18i/21i-B is ".P-L".

Peripheral axis control 1 to 3 may have the extension "P-n" when reading, depending on the CNC model and software version. (Example: When using peripheral axis control 1 to 3 in 1-path system, the extension of peripheral axis control 1 to 3 will be "P-2" to "P-4".) Similarly, even if the file with extension ".PAn" exists in the "storage folder" when writing, it may not be recognized as a standard file name and cannot be written. In this case, when you click the [Show list] / [Execution] button on the startup screen, a message will be displayed to the effect that files with the extension ".PAn" cannot be written. When writing to CNC, change the extension to ".P-n" or use a definition file.

2 If the target CNC is FS16i/18i/21i-B, the dedicated file search and definition file check are performed. Therefore, operate this tool while connected to the CNC.

Default file name (Application file and SRAM data)

When reading from the CNC, the default file names of the application files and SRAM data are as follows.

Table 4.2 (c) List of Default file name (other than FS16*i*/18*i*/21*i*-B)

No.	Default file name	Data kind	Remarks
1	SRAM_BAK.001	SRAM data	Read only
2	OPRM_INF.000	Option information	
3	PMCn.000	PMC sequence program	n: Path number
4	PMCS.000	PMC sequence program for Dual Check Safety	
5	MnPMCMSG	PMC multi-language message	n: Path number
6	CEX_nnM.000	C-language executer application data	nn: Either 05, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90 or A0
7	CEXn.000	C-language executer user data file	n: numeric character
8	PDnn.000	Macro executer application data	n: P-CODE number At FS0*i*-D/PD/Mate-D it will be PCnn.000 At Power Motion *i*-A it will be PEnn.000
9	FPF_20M.000	FANUC PICTURE application data	
10	FPF0.000	FANUC PICTURE screen data	
11	EXSFTKEY.000	Machine operation menu data	
12	LEARN.000	Learning control data	
13	PARTS_LN.000	Learning control for parts cutting data	
14	CMBUSR.000	Customer's board user data	
15	HSCx.000	High-speed cycle cutting data	x: Either N, A, B, C or D
16	FINE_TRQ.000	Fine torque sensing data	
17	IOCONF.000	I/O configuration	
18	RTAP_LN.000	Learning Control for rigid Tapping data	
19	TDG-DATA.000	Machine alarm diagnosis data	
20	ATn_BKUP.000	Automatic data backup data	n:1~3
21	MGICCT01.000	MGi cutting condition data	
22	AITDCINF.000	AI thermal displacement compensation model file	
23	TPRG_DAT.000	Teaching program function	Power Motion *i*-A Plus only
24	LnxxPMCn	Divided ladder program	n: PMC path number xx: 01~99

Table 4.2 (d) List of Default file name (FS16*i*/18*i*/21*i*-B)

No.	Default file name	Data kind	Remarks
1	SRAMxxx☐.FDB	SRAM data	Read only xxx = Symbol indicating the amount of SRAM to be implemented ☐ = Serial number in 512KB units. They are assigned in order from A. If it exceeds 512KB, the SRAM will be split and output as a file, but the files cannot be selected individually.
2	PMC-xxxx.000	Ladder software (for PMC-SB7, PMC-SA1)	x = One alphabetic character
3	PMC@xxxx.000	Ladder software (for loader control)	x = One alphabetic character
4	USR1xxxx.000	Application for PMC C language	x = One alphabetic character
5	MINFO.000	Maintenance information data file	
6	PDnxssss.000	Macro executer/OMM	n = 1/2 (paths) x = M / T (model) ssss = size
7	PDnx_CAP.000	First module for Super CAP	n = 1/2 (paths)

No.	Default file name	Data kind	Remarks
	P1nxssss.000 P1nxssss.000	2nd module for Super CAP 3rd module for Super CAP	x = M / T (model) ssss = size
8	CEX ssss.000	C-language executer(Program)	ssss = size
9	CEXnxxxx.000	C-language executer(data)	n = One–digit number x = One alphabetic character
10	INMC.000	Embedded macro file	

CAUTION
1 The kinds of read user files are depended on the option setting.
2 Write (restore) to SRAM using the boot function (SRAM data cannot be restored while the system is booting).
3 When CNC is operated while reading SRAM data, the contents of SRAM data are not guaranteed even if the tool completes normally.
4 If alarm PS5523 “OPTION AUTHENTICATION WAIT STATE” is output, “OPRM_INF.000” cannot be read and an error will occur.

4.3 DEFINITION FILE

The “definition file” is a file that specifies how to write the file to the CNC. For example, specify in which folder on the CNC the program should be written.

NOTE
The definition file extension is ini. However, this tool uses CNCDataManagementTool.ini and cannot be used as a definition file.

The “definition file” consists of "section", "key", and "value" as shown below. Create the “definition file” (* .INI) according to the file to write. Refer to the "SECTION NAME" section for details.

```
[section]
Key1=value1
Key2=value2
:
```

Below are five representative examples.

[InputFolder]

```
Folder=c:\Machine1\
```

Folder

Specify the folder to store the file to be written to the CNC. Specify an absolute path or a relative path. If not specified, the folder selected in "Select storage folder" (Fig. 3.1 (a)) on the startup screen is treated as the “storage folder”. If relative path is specified, the starting point will be the folder where the definition file exists.

Example)

In the following folder structure, explain the case of writing the files in the storage folder using definition file (Sample.ini).

- Folder structure

```
C:                                  …Drive
└──machine1
    ├──DEFINITION_FILE
    │       Sample.ini              …Definition file
    │
    └──STORAGE_FOLDER               …Storage folder
            CNC-PARA.TXT            …Below, various files
            TOOLOFST.TXT
            TOOLOFST.P-2
```

- How to specify storage folder [InputFolder] in the definition file
  It can be specified as (1) or (2) below.

(1).Relative path specification <7>
Folder=..\STORAGE_FOLDER or Folder=.\..\STORAGE_FOLDER <7>
(2).Absolute path specification <7>
Folder=C:\machine1\STORAGE_FOLDER <7>

[NotWriteFile]

```
File1=prg1.txt
File2=prg2.txt
:
```

File

Specify a file that exists in the “storage folder” but is not to be written. Add a key when there are multiple files. In this case, add the number to the key. If not specified, it is not excluded.

-Specification example

```
[Program]
Folder=//CNC_MEM/USER/PATH1/
File=p*.txt
```

If the “storage folder” has the following files, the File key is "p * .txt", so the following 3 is written. However, prg1.txt and prg2.txt are specified in [NotWriteFile], so only prg3.txt is written.

- prg1.txt …Not Write
- prg2.txt …Not Write
- prg3.txt …Write

[ApplicationFile]

```
File1=P-code1.mem
File2=P-code2.mem
File3=CEXEC1.mem
:
```

File

Specify the user application file created by the macro executor or the C language executor, etc. Add a key when there are multiple files. In this case, add a number to the key. If not specified, it is not written.

[Program]

```
Folder=//CNC_MEM/USER/PATH1/
File=PROG1.TXT
```

Folder

Specify the folder of CNC to write the program. If not specified, clicking the [Show list] / [Execution] button on the startup screen (Fig. 3.1 (a)) displays an error message. <5>

File

Specify the file to write. If not specified, it is not written.

-Program files with folders

If a folder is specified in the program file, it will be written to that folder.

Contents of PROG1.TXT

```
%
&F=//CNC_MEM/
&F=/USER/LIBRARY/
O0001
:
%
```

Even if specify to write PROG1.TXT to "//CNC_MEM/USER/PATH1/" with "definition file" as shown below, PROG1.TXT is written to "//CNC_MEM/USER/LIBRARY/".

Folder=//CNC_MEM/USER/PATH1/
File=PROG1.TXT

-File name

Wildcards can be used in each file name.

Folder=//CNC_MEM/USER/PATH1/
File=PROG1*.*

In the above case, the files starting with "PROG1" are targeted.

- PROG11_01.TXT …Write
- PROG12_02.txt …Write
- PROG21_01.txt …Not Write

-File name with folder specification

By specifying a folder for the file name, files stored in folders other than [Input folder] can be written.

File=c:\common\CMN-PROG.TXT CMN-PROG.TXT stored in c:\common\ is written.

[ToolOffsetData]

Path=1
File=TOOLOFS1.TXT

Path

Specify the path of the CNC to write the offset data. If not specified, clicking the [Show list] / [Execution] button on the startup screen (Fig.3.1 (a)) displays an error message. <5>

File

Specify the file to write. If not specified, it is not written.

NOTE

1 Except for the [NotWriteFile] section and the [ApplicationFile] section, only one key can be specified in one section. Therefore, when writing multiple files of the same type such as a program, specify the sections separately. In this case, add a number to the section name.

Example 1) When writing the program files PROG11.TXT and PROG12.TXT to //CNC_MEM/USER/PATH1/

[Program1]
Folder=//CNC_MEM/USER/PATH1/
File=PROG11.TXT
[Program2]
Folder=//CNC_MEM/USER/PATH1/
File=PROG12.TXT

Example 2) When writing tool offset data TOOLOFS1.TXT to Path 1 and TOOLOFS2.TXT to Path 2

[ToolOffsetData1]
Path=1
File=TOOLOFS1.TXT
[ToolOffsetData2]
Path=2
File=TOOLOFS2.TXT

NOTE

Example 3)

Specify the path number including the machining path when writing to the loader path in the system where the loader control function is enabled. When the system is composed of machining path1, machining path2 and loader path1, specify as follows in order to write MACRO_P1.TXT to machining path1, MACRO_P2.TXT to machining path2, and MACRO_LC1.TXT to loader path1. <6>

```
[CustomMacroCommonVariable1]
  Path=1
  File=MACRO_P1.TXT
[CustomMacroCommonVariable2]
  Path=2
  File=MACRO_P2.TXT
[CustomMacroCommonVariable3]
  Path=3
  File=MACRO_LC1.TXT
```

<6>

2 Up to 1000 numbers can be added to the "section" and "key". In the following cases, "File1001 = CEXEC1.mem" will not be written.

```
[ApplicationFile]
File1=P-code1.mem
File2=P-code2.mem
:
File1001=CEXEC1.mem
```

3 All FS16*i*/18*i*/21*i*-B text files are the path data file. The explanation method is the same as [ToolOffsetData] explained in the example. <8>

<8>

The same applies to other files.

4.4 SECTION NAME

The sections specified in the “definition file” are as follows. Also, functions that can only be read cannot be specified in the definition file, but they are output to the log with the following section names. <5>

<5>

Table 4.4 (a) Section

[InputFolder]	Storage folder
[NotWriteFile]	File not written
[ApplicationFile]	Application file
[Program]	Program
[ToolOffsetData]	Tool offset
[Parameter]	Parameter
[PitchErrorCompensationData]	Pitch error compensation data
[3-DimensionalErrorCompensationData]	Three-dimensional error compensation data
[3-DimensionalRotaryErrorCompensationData]	Three-dimensional rotary error compensation data <2>
[CustomMacroCommonVariable]	Custom macro common variable
[WorkpieceCoordinateSystemData]	Workpiece coordinate system data
[ToolMmanagementData]	Tool management data (tool management function)
[MagazineData]	Magazine data (tool management function)
[ToolLifeStatusName]	Tool life status name (tool management function)
[CustomizeDataName]	Customize data name (tool management function)
[CustomizeDataDisplayed]	Customize data displayed as tool management data (tool management expansion)
[SpindleWaitingPositionName]	Spindle waiting position name (tool management expansion)
[DecimalPointPositionData]	Decimal point position data of customize data (tool management expansion)
[ToolGeometryData]	Tool geometry data (tool management function for oversize tools)
[WorkpieceSettingErrorValue]	Workpiece setting error value
[ToolLifeManagementData]	Tool life management data

[MaintenanceInformation]	Maintenance information	
[PeriodicMaintenanceData]	Periodic maintenance data (periodic maintenance: status screen)	
[SecondGeometryToolOffset]	Second geometry tool offset	
[RotaryTableDynamicFixtureOffset]	Rotary table dynamic fixture offset	
[RealTimeCustomMacroVariable]	Real time custom macro variable	
[PMCSignalProtect]	PMC signal protect	
[CauseData]	Cause data (tool management tool attachment/detachment function)	
[MachineSystemNameData]	Machine system name data (periodic maintenance: machine system screen)	
[PCodeVariable]	P code variable	
[MCodeGroupData]	M code group data	
[ToolGeometrySizeData]	Tool geometry size data	
[CustomersBoard]	Customer’s board	
[OperationHistorySignalSectionData]	Operation history signal section data	
[3DInterferenceCheckData]	3D interference check data	
[AllToolData]	All tool data (Tool management function)	
[AllCustomData]	All custom data (Tool management function)	
[MachineConfiguration]	Machine configuration	
[MachiningConditionData]	Machining condition data	<9>
[ScrollWaitingMcode]	Scroll waiting M-code	<9>
[TableDataOfSmartSpindleLoadControl]	Table data of smart spindle-load control	<9>
[WorkpieceOriginShift]	Workpiece coordinate system shift	<10>
[SoftOpPanel]	Software operator's panel	<10>
[SoftOpPanelGeneralPurpose]	Software operator's panel general-purpose switch	<10>
[ToolData]	Tool data	<4>
[CuttingConditionData]	Cutting condition data	<4>
[PMCParameter]	PMC parameter	
[DCSafetyPMCParameter]	PMC parameter (Dual Check Safety)	

The following are read only functions. <5>

[SramData]	SRAM data	<6>
[SystemConfigurationData]	System configuration data	<5>
[OperationHistoryData]	Operation history data	
[CncIdInformation]	CNC ID information	
[SystemAlarmInformation]	System alarm information	
[ServoSpindleInformation]	Servo/spindle information	
[AttachmentHistoryData]	Attachment history data (tool management tool attachment / detachment function)	
[DetachmentHistoryData]	Detachment history data (tool management tool attachment / detachment function)	
[MachineStateHistoryData]	Machine state history data	
[PowerCalibrationCoefficientHistory]	Power calibration coefficient history	
[PartsRunHour]	Parts run hour	
[HistoryOfTheAlarmRelatedToDischarge]	History of the alarm related to discharge	
[LaserAlarmHistory]	Laser alarm history	
[WarmUpHistory]	Warm up history	<5>
[DiagnosisData]	Diagnosis data	<8>
[iHMIVersionInformation]	iHMI version information	<9>
[PathTableOperationHistoryFunction]	Path table operation history function	
[OperationHistoryBackedUpInFrom]	Operation history backed up in FROM	
[AutoMeasurementData]	Auto measurement data	
[LaserTrendData]	Laser trend data	<9>
[EnabledFunction]	Enabled function	<13>

4.5 CORRESPONDENCE BY MODEL

The files that can be write / read differ depending on the model. See the table below. Even if unsupported data is specified in the definition file, it will not be write.

W : Writable
R : Readable
－ : Not writable / readable

Table 4.5 (a) Correspondence table for each model (application file)

Function	30*i*/31*i*/32*i*-MODEL A STEP2,3	30*i*/31*i*/32*i*-MODEL A STEP4	30*i*/31*i*/32*i*-MODEL B STEP1	30*i*/31*i*/32*i*-MODEL B STEP2	30*i*/31*i*/32*i*-MODEL B Plus	30*i*-MODEL B STEP1 (Multi axes)	35*i*-MODEL B STEP1	35*i*-MODEL B STEP2	Power Motion *i*-MODEL A	Power Motion *i*-MODEL A Plus
SRAM data(*1)	R/-(B)	R/-(B)	R/-(B)	R/-(A)	R/-(A)	R/-(B)	R/-(B)	R/-(A)	R/-(B)	R/-(A)
Option information	－	－	R/-	R/-	R/-	－	R/-	R/-	R/-	R/-
PMC sequence program	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
PMC sequence program for Dual Check Safety	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
PMC multi-language message	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
I/O configuration	－	－	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
C-language executer application data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
C-language executer user data file	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Macro executer application data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
FANUC PICTURE application data	R/-	R/-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
FANUC PICTURE screen data	R/-	R/-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Machine operation menu data	R/-	R/-	R/-	R/W	R/W	R/-	－	－	－	R/W
Customer's board user data	R/-	R/-	R/-	R/W	R/W	－	－	－	－	－
Machine alarm diagnosis data	R/-	R/-	R/W	R/W	R/W	R/-	R/W	R/W	R/W	R/W
Learning control data	R/-	R/-	R/-	R/-	R/-	－	－	－	－	－
Learning control for parts cutting data	R/-	R/-	R/-	R/-	R/-	－	－	－	－	－
Learning Control for rigid Tapping data	R/-	R/-	R/-	R/-	R/-	－	－	－		－
High-speed cycle cutting data	R/W	R/W	R/W	R/W	R/W	－	－	－	－	－
Fine torque sensing data	R/W	R/W	R/W	R/W	R/W	－	－	－	－	－
Automatic data backup data	－	－	R/W	R/W	R/W	－	－	R/W	－	R/W
MGi cutting condition data	－	－	R/W	R/W	R/W	－	－	R/W	－	－
AI thermal displacement compensation model file	－	－	－	R/W	R/W	－	－	R/W	－	－
Teaching program function	－	－	－	－	－	－	－	－	R/W	R/W
Divided ladder program	－	－	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Function	0*i*-MODEL D 0*i* Mate-MODEL D	0*i*-MODEL F STEP1	0*i*-MODEL F STEP2	0*i*-MODEL F Plus STEP1, 2
SRAM data(*1)	R/-(B)	R/-(B)	R/-(A)	R/-(A)
Option information	－	R/-	R/-	R/-

Function	0*i*-MODEL D 0*i* Mate-MODEL D	0*i*-MODEL F STEP1	0*i*-MODEL F STEP2	0*i*-MODEL F Plus STEP1, 2
PMC sequence program	R/W	R/W	R/W	R/W
PMC sequence program for Dual Check Safety	R/W	R/W	R/W	R/W
PMC multi-language message	R/W	R/W	R/W	R/W
I/O configuration	—	R/W	R/W	R/W
C-language executer application data	R/W	R/W	R/W	R/W
C-language executer user data file	R/W	R/W	R/W	R/W
Macro executer application data	R/W	R/W	R/W	R/W
FANUC PICTURE application data	R/W	R/W	R/W	R/W
FANUC PICTURE screen data	R/W	R/W	R/W	R/W
Machine operation menu data	R/W	R/-	R/W	R/W
Customer's board user data	—	—	—	—
Machine alarm diagnosis data	R/W	R/W	R/W	R/W
Learning control data	—	—	—	—
Learning control for parts cutting data	—	—	—	—
Learning Control for rigid Tapping data	—	—	—	—
High-speed cycle cutting data	—	—	—	—
Fine torque sensing data	—	—	—	—
Automatic data backup data	—	—	R/W	R/W
MGi cutting condition data	—	—	R/W	R/W
AI thermal displacement compensation model file	—	—	R/W	R/W
Teaching program function	—	—	—	—
Divided ladder program	—	R/W	R/W	R/W

Function	16*i*/18*i*/21*i*-MODEL B
SRAM data	R/- (A)
Ladder software (for PMC-SB7, PMC-SA1)	R/W
Ladder software (for loader control)	R/W
PMC C language application	R/W
Maintenance information data file	R/W
Macro executer/OMM (order made macro)	R/W
First module for Super CAP Second module for Super CAP Third module for Super CAP	R/W
C-language executer (Program)	R/W
C-language executer (data)	R/W
Embedded macro file	R/W

Function	31*i*-WA	30*i*/31*i*-LB STEP1	30*i*/31*i*-LB STEP2	30*i*/31*i*-PB STEP1	30*i*/31*i*-PB STEP2	31*i*-PB Plus	31*i*-WB STEP1	31*i*-WB STEP2	30*i*/31*i*-LB Plus	LSC*i*-A
SRAM data(*1)	R/-(B)	R/-(B)	R/-(A)	R/-(B)	R/-(A)	R/-(A)	R/-(B)	R/-(A)	R/-(A)	R/-(A)
Option information	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
PMC sequence program	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
PMC sequence program for Dual Check Safety	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
PMC multi-language message	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
I/O configuration	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
C-language executer application data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
C-language executer user data file	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Macro executer application data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
FANUC PICTURE application data	R/-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
FANUC PICTURE screen data	R/-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Machine operation menu data	R/-	—	R/W	—	R/W	R/W	R/-	R/W	R/W	R/W
Customer's board user data	—	—	R/W	—	—	—	—	—	R/W	—
Machine alarm diagnosis data	R/-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Learning control data	—	R/-	R/-	R/-	R/-	R/-	—	—	R/-	R/-
Learning control for parts cutting data	—	—	—	—	—	—	—	—	—	—
Learning control for rigid tapping data	—	—	—	—	—	—	—	—	—	—
High-speed cycle cutting data	—	—	—	—	—	—	—	—	—	—
Fine torque sensing data	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	R/W
Automatic data backup data	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
MGi cutting condition data	—	—	—	—	—	—	—	—	—	—
AI thermal displacement compensation model file	—	—	—	—	—	—	—	—	—	—
Teaching program function	—	—	—	—	—	—	—	—	—	—
Divided ladder program	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Function	0*i*-P-D	0*i*-PF STEP1	0*i*-PF STEP2	0*i*-LF Plus	0*i*-PF Plus
SRAM data(*1)	R/-(B)	R/-(B)	R/-(A)	R/-(A)	R/-(A)
Option information	—	R/-	R/-	R/-	R/-
PMC sequence program	R/W	R/W	R/W	R/W	R/W
PMC sequence program for Dual Check Safety	R/W	R/W	R/W	R/W	R/W
PMC multi-language message	R/W	R/W	R/W	R/W	R/W
I/O configuration	—	R/W	R/W	R/W	R/W
C-language executer application data	R/W	R/W	R/W	R/W	R/W
C-language executer user data file	R/W	R/W	R/W	R/W	R/W
Macro executer application data	R/W	R/W	R/W	R/W	R/W

<14>

Function	0*i*-P-D	0*i*-PF STEP1	0*i*-PF STEP2	0*i*-LF Plus	0*i*-PF Plus
FANUC PICTURE application data	R/W	R/W	R/W	R/W	R/W
FANUC PICTURE screen data	R/W	R/W	R/W	R/W	R/W
Machine operation menu data	R/W	—	R/W	R/W	R/W
Customer's board user data	—	—	—	—	—
Machine alarm diagnosis data	R/W	R/W	R/W	R/W	R/W
Learning control data	—	—	—	—	—
Learning control for parts cutting data	—	—	—	—	—
Learning Control for rigid Tapping data	—	—	—	—	—
High-speed cycle cutting data	—	—	—	—	—
Fine torque sensing data	—	—	—	—	—
Automatic data backup data	—	R/W	R/W	R/W	R/W
MGi cutting condition data	—	—	—	—	—
AI thermal displacement compensation model file	—	—	—	—	—
Teaching program function	—	—	—	—	—
Divided ladder program	—	R/W	R/W	R/W	R/W

<14>
<14>

(*1) The model (A) can be read even when the emergency stop is released. The model (B) can be read only in the emergency stop state.

<7>

Table 4.5 (b) Correspondence table for each model (text file)

<12>

Function	30*i*/31*i* 32*i*-MODEL A STEP2,3	30*i*/31*i*/32*i*-MODEL A STEP4	30*i*/31*i*/32*i*-MODEL B STEP1	30*i*/31*i*/32*i*-MODEL B STEP2	30*i*/31*i*/32*i*-MODEL B Plus	30*i*-MODEL B STEP1(Multi axes)	35*i*-MODEL B STEP1	35*i*-MODEL B STEP2	Power Motion *i*-MODEL A	Power Motion *i*-MODEL A Plus	16*i*/18*i*/21*i*-MODEL B
Program	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Tool offset	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Parameter	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Pitch error compensation data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
3-dimensional error compensation data	—	—	R/-	R/W	R/W	—	—	—	—	—	—
3-dimensional rotary error compensation data	—	R/W	R/W	R/W	R/W	R/W	—	—	—	—	—
Custom macro common variable	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Workpiece coordinate system data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Tool management data (tool management function)	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—
Magazine data (tool management function)	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—
Tool life status name (tool management function)	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—
Customize data name (tool management function)	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—

<12>

Function	30*i*/31*i*/ 32*i*-MODEL A STEP2,3	30*i*/31*i*/32*i*-MODEL A STEP4	30*i*/31*i*/32*i*-MODEL B STEP1	30*i*/31*i*/32*i*-MODEL B STEP2	30*i*/31*i*/32*i*-MODEL B Plus	30*i*-MODEL B STEP1(Multi axes)	35*i*-MODEL B STEP1	35*i*-MODEL B STEP2	Power Motion *i*-MODEL A	Power Motion *i*-MODEL A Plus	16*i*/18*i*/21*i*-MODEL B	
Customize data displayed as tool management data	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—	<12>
Spindle waiting position name	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—	
Decimal point position data of customize data	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—	
Tool geometry data	R/W	R/W	R/W	R/W	R/W	—	—	R/W	—	—	—	
Workpiece setting error value	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	—	—	
Tool life management data	—	—	R/W	R/W	R/W	—	—	—	—	—	—	
Maintenance information	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	
Periodic maintenance data	—	—	R/-	R/W	R/W	—	—	R/W	—	R/W	—	
Second geometry tool offset	—	—	R/-	R/W	R/W	—	—	—	—	—	—	
Rotary table dynamic fixture offset	R/-	—	R/W	R/W	R/W	—	R/W	R/W	—	—	—	
Real time custom macro variable	—	—	R/-	R/W	R/W	—	—	R/W	—	—	—	
PMC signal protect	—	—	R/-	R/W	R/W	—	—	R/W	—	—	—	
Cause data	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W	—	—	—	
Machine system name data	—	—	R/-	R/W	R/W	—	—	R/W	—	R/W	—	
P code variable	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
M code group data	—	R/W	R/W	R/W	R/W	R/W	—	R/W	—	—	R/W (*3)	
Tool geometry size data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	—	—	
Customer's board	—	—	R/-	R/W	R/W	—	—	—	—	—	—	
Operation history signal section data	—	-/W	R/-	R/W	R/W	—	—	R/W	—	R/W	—	
3D interference check data	R/W	R/W	R/W	R/W	R/W	—	—	—	—	—	—	
All tool data	—	—	R/-	R/W	R/W	—	—	R/W	—	—	—	
All custom data	—	—	R/-	R/W	R/W	—	—	R/W	—	—	—	
Machine configuration	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	—	—	
Machining condition data	—	—	R/W	R/W	R/W	—	—	—	—	—	—	
Scroll waiting M-code	—	—	R/W	R/W	R/W	—	—	—	—	—	—	
Table data of smart spindle-load control	—	—	R/W	R/W	R/W	—	—	—	—	—	—	
Workpiece coordinate system shift	R/W	R/W	R/W	R/W	R/W	R/W	—	—	—	—	R/W	
Software operator's panel	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Software operator's panel general-purpose switch	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Tool data	—	—	—	—	—	—	—	—	—	—	—	<12>
Cutting condition data	—	—	—	R/W (*4)	R/W (*4)	—	—	—	—	—	—	<14> <14>
PMC parameter	—	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	<12>
PMC parameter (Dual Check Safety)	—	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—	
CNC ID information	R/- (*2)	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	—	
System alarm information	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	—	
Servo/spindle information	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	—	
System configuration data	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	—	<12>

Function	30*i*/31*i* 32*i*-MODEL A STEP2,3	30*i*/31*i*/32*i*-MODEL A STEP4	30*i*/31*i*/32*i*-MODEL B STEP1	30*i*/31*i*/32*i*-MODEL B STEP2	30*i*/31*i*/32*i*-MODEL B Plus	30*i*-MODEL B STEP1(Multi axes)	35*i*-MODEL B STEP1	35*i*-MODEL B STEP2	Power Motion *i*-MODEL A	Power Motion *i*-MODEL A Plus	16*i*/18*i*/21*i*-MODEL B	
Operation history data	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	<12>
Attachment history data	R/-	R/-	R/-	R/-	R/-	—	R/-	R/-	—	—	—	<12>
Detachment history data	R/-	R/-	R/-	R/-	R/-	—	R/-	R/-	—	—	—	<12>
Machine state history data	—	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	—	<12>
Power calibration coefficient history	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	<14>
Parts run hour	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	
History of the alarm related to discharge	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	
Laser alarm history	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	
Warm up history	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	<14>
Diagnosis data	—	—	R/-	R/-	R/-	R/-	—	R/-	—	R/-	—	<12>
iHMI version information	—	—	—	R/-	R/-	—	—	—	—	R/-	—	<12>
Path table operation history function	—	R/-	R/-	R/-	R/-	R/-	—	—	—	—	—	<12>
Operation history backed up in FROM	—	—	—	R/-	R/-	—	—	R/-	—	R/-	—	<12>
Auto measurement data	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	<14> <14>
Laser trend data	—	—	—	R/- (*4)	R/- (*4)	—	—	—	—	—	—	<14> <14>
Enabled function	—	—	—	R/-	R/-	—	—	R/-	—	R/-	—	<13>

(*2) Only for 30i/31i/32i-MODEL A STEP3 <7>

(*3) Cannot be read while in the emergency stop state <8>

(*4) Only for laser control function <14>

Function	0*i*-MODEL D 0*i* Mate-MODEL D	0*i*-MODEL F STEP1	0*i*-MODEL F STEP2	0*i*-MODEL F Plus STEP1, 2	
Program	R/W	R/W	R/W	R/W	<12>
Tool offset	R/W	R/W	R/W	R/W	
Parameter	R/W	R/W	R/W	R/W	
Pitch error compensation data	R/W	R/W	R/W	R/W	
3-dimensional error compensation data	—	—	—	—	
3-dimensional rotary error compensation data	—	—	—	—	
Custom macro common variable	R/W	R/W	R/W	R/W	<12>

Function	0*i*-MODEL D 0*i* Mate-MODEL D	0*i*-MODEL F STEP1	0*i*-MODEL F STEP2	0*i*-MODEL F Plus STEP1, 2
Workpiece coordinate system data	R/W	R/W	R/W	R/W
Tool management data (tool management function)	R/W	R/W	R/W	R/W
Magazine data (tool management function)	R/W	R/W	R/W	R/W
Tool life status name (tool management function)	R/W	R/W	R/W	R/W
Customize data name (tool management function)	R/W	R/W	R/W	R/W
Customize data displayed as tool management data	R/W	R/W	R/W	R/W
Spindle waiting position name	—	R/W	R/W	R/W
Decimal point position data of customize data	R/W	R/W	R/W	R/W
Tool geometry data	—	R/W	R/W	R/W
Workpiece setting error value	—	—	—	—
Tool life management data	—	—	R/W	—
Maintenance information	R/W	R/W	R/W	R/W
Periodic maintenance data	-/W	R/-	R/W	R/W
Second geometry tool offset	—	R/-	R/W	R/W
Rotary table dynamic fixture offset	—	—	—	—
Real time custom macro variable	—	R/-	R/W	R/W
PMC signal protect	—	R/-	R/W	R/W
Cause data	R/W	R/W	R/W	R/W
Machine system name data	-/W	R/-	R/W	R/W
P code variable	R/W	R/W	R/W	R/W
M code group data	—	R/W	R/W	R/W
Tool geometry size data	—	R/W	R/W	R/W
Customer's board	—	—	—	—
Operation history signal section data	-/W	R/-	R/W	R/W
3D interference check data	—	—	—	—
All tool data	-/W	R/-	R/W	R/W
All custom data	-/W	R/-	R/W	R/W
Machine configuration	—	—	—	—
Machining condition data	—	—	—	—
Scroll waiting M-code	—	R/W	R/W	R/W
Table data of smart spindle-load control	—	R/W	R/W	R/W
Workpiece coordinate system shift	R/W	R/W	R/W	R/W
Software operator's panel	R/W	R/W	R/W	R/W
Software operator's panel general-purpose switch	R/W	R/W	R/W	R/W
Tool data	—	—	—	—
Cutting condition data	—	—	—	—
PMC parameter	R/W	R/W	R/W	R/W
PMC parameter (Dual Check Safety)	R/W	R/W	R/W	R/W
CNC ID information	R/-	R/-	R/-	R/-
System alarm information	R/-	R/-	R/-	R/-

Function	0*i*-MODEL D 0*i* Mate-MODEL D	0*i*-MODEL F STEP1	0*i*-MODEL F STEP2	0*i*-MODEL F Plus STEP1, 2
Servo/spindle information	R/-	R/-	R/-	R/-
System configuration data	R/-	R/-	R/-	R/-
Operation history data	R/-	R/-	R/-	R/-
Attachment history data	R/-	R/-	R/-	R/-
Detachment history data	R/-	R/-	R/-	R/-
Machine state history data	—	R/-	R/-	R/-
Power calibration coefficient history	—	—	—	—
Parts run hour	—	—	—	—
History of the alarm related to discharge	—	—	—	—
Laser alarm history	—	—	—	—
Warm up history	—	—	—	—
Diagnosis data	—	R/-	R/-	R/-
iHMI version information	—	—	—	R/-
Path table operation history function	—	—	—	—
Operation history backed up in FROM	—	—	R/-	R/-
Auto measurement data	—	—	—	—
Laser trend data	—	—	—	—
Enabled function	—	—	R/-	R/-

Function	31*i*-WA	30*i*/31*i*-LB STEP1	30*i*/31*i*-LB STEP2	30*i*/31*i*-PB STEP1	30*i*/31*i*-PB STEP2	31*i*-PB Plus	31*i*-WB STEP1	31*i*-WB STEP2	30*i*/31*i*-LB Plus	LSC*i*-A
Program	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Tool offset	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Parameter	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Pitch error compensation data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
3-dimensional error compensation data	—	R/-	R/W	R/-	R/W	R/W	R/-	R/W	R/W	—
3-dimensional rotary error compensation data	—	—	—	—	—	—	R/W	R/W	—	—
Custom macro common variable	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Workpiece coordinate system data	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Tool management data (tool management function)	—	—	—	—	—	—	—	—	—	—
Magazine data (tool management function)	—	—	—	—	—	—	—	—	—	—
Tool life status name (tool management function)	—	—	—	—	—	—	—	—	—	—
Customize data name (tool management function)	—	—	—	—	—	—	—	—	—	—
Customize data displayed as tool management data	—	—	—	—	—	—	—	—	—	—
Spindle waiting position name	—	—	—	—	—	—	—	—	—	—

Function	31*i*-WA	30*i*/31*i*-LB STEP1	30*i*/31*i*-LB STEP2	30*i*/31*i*-PB STEP1	30*i*/31*i*-PB STEP2	31*i*-PB Plus	31*i*-WB STEP1	31*i*-WB STEP2	30*i*/31*i*-LB Plus	LSC*i*-A
Decimal point position data of customize data	—	—	—	—	—	—	—	—	—	—
Tool geometry data	—	—	—	—	—	—	—	—	—	—
Workpiece setting error value	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	—
Tool life management data	—	—	—	—	—	—	—	—	—	—
Maintenance information	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Periodic maintenance data	—	R/-	R/W	R/-	R/W	R/W	R/-	R/W	R/W	R/W
Second geometry tool offset	—	—	—	—	—	—	—	—	—	—
Rotary table dynamic fixture offset	—	—	—	—	—	—	R/W	R/W	—	—
Real time custom macro variable	—	R/-	R/W	R/-	R/W	R/W	—	—	R/W	R/W
PMC signal protect	—	R/-	R/W	R/-	R/W	R/W	—	—	R/W	R/W
Cause data	—	—	—	—	—	—	—	—	—	—
Machine system name data	—	R/-	R/W	R/-	R/W	R/W	R/-	R/W	R/W	R/W
P code variable	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
M code group data	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	R/W
Tool geometry size data	—	—	—	—	—	—	—	—	—	—
Customer's board	—	—	R/W	—	—	—	—	—	R/W	—
Operation history signal section data	—	R/-	R/W	R/-	R/W	R/W	R/-	R/W	R/W	R/W
3D interference check data	—	—	—	—	—	—	—	—	—	—
All tool data	—	—	—	—	—	—	—	—	—	—
All custom data	—	—	—	—	—	—	—	—	—	—
Machine configuration	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	—
Machining condition data	—	—	—	—	—	—	—	—	—	—
Scroll waiting M-code	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	—
Table data of smart spindle-load control	—	—	—	—	—	—	—	—	—	—
Workpiece coordinate system shift	—	—	—	—	—	—	—	—	—	—
Software operator's panel	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Software operator's panel general-purpose switch	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Tool data	—	R/W	R/W	R/W	R/W	R/W	—	—	R/W	—
Cutting condition data	—	R/W	R/W	—	—	—	—	—	R/W	R/W
PMC parameter	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
PMC parameter (Dual Check Safety)	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
CNC ID information	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
System alarm information	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
Servo/spindle information	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
System configuration data	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
Operation history data	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
Attachment history data	—	—	—	—	—	—	—	—	—	—
Detachment history data	—	—	—	—	—	—	—	—	—	—
Machine state history data	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
Power calibration coefficient history	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Parts run hour	—	R/-	R/-	—	—	—	—	—	R/-	R/-
History of the alarm related to discharge	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Laser alarm history	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Warm up history	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Diagnosis data	—	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-	R/-
iHMI version information	—	—	R/-	—	R/-	R/-	—	R/-	R/-	—

Function	31*i*-WA	30*i*/31*i*-LB STEP1	30*i*/31*i*-LB STEP2	30*i*/31*i*-PB STEP1	30*i*/31*i*-PB STEP2	31*i*-PB Plus	31*i*-WB STEP1	31*i*-WB STEP2	30*i*/31*i*-LB Plus	LSC*i*-A
Path table operation history function	—	—	—	—	—	—	—	—	—	—
Operation history backed up in FROM	—	—	R/-	—	R/-	R/-	—	R/-	R/-	R/-
Auto measurement data	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Laser trend data	—	R/-	R/-	—	—	—	—	—	R/-	R/-
Enabled function	—	—	R/-	—	R/-	R/-	—	R/-	R/-	R/-

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Function	0*i*-PD	0*i*-PF STEP1	0*i*-PF STEP2	0*i*-LF Plus	0*i*-PF Plus
Program	R/W	R/W	R/W	R/W	R/W
Tool offset	R/W	R/W	R/W	R/W	R/W
Parameter	R/W	R/W	R/W	R/W	R/W
Pitch error compensation data	R/W	R/W	R/W	R/W	R/W
3-dimensional error compensation data	—	—	—	—	—
3-dimensional rotary error compensation data	—	—	—	—	—
Custom macro common variable	R/W	R/W	R/W	R/W	R/W
Workpiece coordinate system data	R/W	R/W	R/W	R/W	R/W
Tool management data (tool management function)	—	—	—	—	—
Magazine data (tool management function)	—	—	—	—	—
Tool life status name (tool management function)	—	—	—	—	—
Customize data name (tool management function)	—	—	—	—	—
Customize data displayed as tool management data	—	—	—	—	—
Spindle waiting position name	—	—	—	—	—
Decimal point position data of customize data	—	—	—	—	—
Tool geometry data	—	—	—	—	—
Workpiece setting error value	—	—	—	—	—
Tool life management data	—	—	—	—	—
Maintenance information	R/W	R/W	R/W	R/W	R/W
Periodic maintenance data	—	R/-	R/W	R/W	R/W
Second geometry tool offset	—	—	—	—	—
Rotary table dynamic fixture offset	—	—	—	—	—
Real time custom macro variable	—	R/-	R/W	R/W	R/W
PMC signal protect	—	R/-	R/W	R/W	R/W
Cause data	—	—	—	—	—
Machine system name data	—	R/-	R/W	R/W	R/W
P code variable	R/W	R/W	R/W	R/W	R/W
M code group data	—	R/W	R/W	R/W	R/W
Tool geometry size data	—	—	—	—	—

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Function	0i-PD	0i-PF STEP1	0i-PF STEP2	0i-LF Plus	0i-PF Plus
Customer's board	—	—	—	—	—
Operation history signal section data	—	R/-	R/W	R/W	R/W
3D interference check data	—	—	—	—	—
All tool data	—	—	—	—	—
All custom data	—	—	—	—	—
Machine configuration	—	—	R/W	R/W	R/W
Machining condition data	—	—	—	—	—
Scroll waiting M-code	—	—	—	—	—
Table data of smart spindle-load control	—	—	—	—	—
Workpiece coordinate system shift	—	—	—	—	—
Software operator's panel	R/W	R/W	R/W	R/W	R/W
Software operator's panel general-purpose switch	R/W	R/W	R/W	R/W	R/W
Tool data	—	R/W	R/W	—	R/W
Cutting condition data	—	—	—	R/W	—
PMC parameter	R/W	R/W	R/W	R/W	R/W
PMC parameter (Dual Check Safety)	R/W	R/W	R/W	R/W	R/W
CNC ID information	R/-	R/-	R/-	R/-	R/-
System alarm information	R/-	R/-	R/-	R/-	R/-
Servo/spindle information	R/-	R/-	R/-	R/-	R/-
System configuration data	R/-	R/-	R/-	R/-	R/-
Operation history data	R/-	R/-	R/-	R/-	R/-
Attachment history data	—	—	—	—	—
Detachment history data	—	—	—	—	—
Machine state history data	—	R/-	R/-	R/-	R/-
Power calibration coefficient history	—	—	—	R/-	—
Parts run hour	—	—	—	R/-	—
History of the alarm related to discharge	—	—	—	R/-	—
Laser alarm history	—	—	—	R/-	—
Warm up history	—	—	—	R/-	—
Diagnosis data	—	R/-	R/-	R/-	R/-
iHMI version information	—	—	—	R/-	—
Path table operation history function	—	—	—	—	—
Operation history backed up in FROM	—	—	R/-	R/-	R/-
Auto measurement data	—	—	—	R/-	—
Laser trend data	—	—	—	R/-	—
Enabled function	—	—	R/-	R/-	R/-

4.5.1 Model and series

The relationship between the model and the series is as follows.

Table 4.5.1 (a) Model and series correspondence table

Model	Series
0*i*-MODEL D 0*i* Mate-MODEL D	D4F1, D6F1 D5F1, D7F1 D4F2, D6F2
0*i*-MODEL F STEP1	D4G1, D6G1
0*i*-MODEL F STEP2	D4G2, D6G2 D5G2, D7G2
0*i*-MODEL F Plus STEP1	D4G3, D6G3
0*i*-MODEL F Plus STEP2	D4G5, D6G5
30*i*/31*i*/32*i*-MODEL A STEP2	G002, G012, G022, G032, G00B, G01B, G02B, G03B G00K, G01K, G02K, G03K, G00T, G01T, G02T, G03T G101, G111 G10K, G11K G121, G131, G12B, G13B G12K, G13K, G12T, G13T G201 G20K
30*i*/31*i*/32*i*-MODEL A STEP3	G003, G013, G023, G033, G00C, G01C, G02C, G03C G00L, G01L, G02L, G03L, G00U, G01U, G02U, G03U G103, G113 G10L, G11L G123, G133, G12C, G13C G12L, G13L, G12U, G13U G203 G20L
30*i*/31*i*/32*i*-MODEL A STEP4	G004, G014, G024, G034 G00D, G01D, G02D, G03D G124, G134, G104, G114 G12D, G13D, G10D, G11D G204 G20D
31*i*-WA	G191, G1BA
30*i*/31*i*/32*i*-MODEL B STEP1	G301, G311, G321, G331, G351 G30A, G31A, G32A, G33A, G35A G421, G431 G42A, G43A, G481 G401, G411 G40A, G41A, G451 G501, G511 G50A, G51A, G521
30*i*/31*i*/32*i*-MODEL B STEP2	G303, G313, G323, G333, G353 G423, G433, G483, G4H3 G403, G413, G453, G4G3 G503, G513, G523 G3L3, G3M3 G4N3, G4P3 G4L3, G4M3 G5L3

Model	Series	
30*i*/31*i*/32*i*-MODEL B Plus	G307, G317, G327, G337, G357, G32D, G35D, G3GD, G3HD G427, G437, G487, G4H7, G4HD G407, G417, G457, G4G7, G4GD G507, G517, G527, G547, G52D G3L7, G3M7, G3MD G4N7, G4P7, G4PD G4L7, G4M7, G4MD G5L7	<9> <9> <9> <7> <14> <14>
30*i*-MODEL B STEP1 (Multi axes)	G3G1, G3H1	<7>
35*i*-MODEL B STEP1	G601, G611	<7>
35*i*-MODEL B STEP2	G603, G613	<7>
31*i*-WB STEP1	G4A1	<12>
31*i*-WB STEP2	G4A4	<12>
LSC*i*-A	G4QD	<14>
Power Motion *i*-MODEL A	88H0	<7>
Power Motion *i*-MODEL A Plus	88H2, 88HD	<12>
0*i*-PD	D8F1, D8F2	<7>
0*i*-PF STEP1	D8G1	<7>
0*i*-PF STEP2	D8G2	<7>
0*i*-PF Plus	D8G5	<14>
0*i*-LF Plus	DCG3, DCG4	<14>
30*i*/31*i*-LB STEP1	G3C1, G3D1, G3C2, G3D2, G3C3, G3D3 G4C1, G4C2, G4C3	<7>
30*i*/31*i*-LB STEP2	G3C4, G3D4, G3C5, G3D5 G4B4, G4C4, G4B5, G4C5	
30*i*/31*i*-PB STEP1	G3E1, G3F1 G4E1	
30*i*/31*i*-PB STEP2	G3E4, G3F4 G4E4	
30*i*/31*i*-LB Plus	G3CE, G3DE, G4BE, G4CE G3CD, G3DD, G4BD, G4CD G3C8, G3D8, G3C7, G3D7 G4B8, G4C8, G4B7, G4C7	<7> <14> <14>
31*i*-PB Plus	G4ED	<13>
16*i*-B	B0H1, B0HA, B0HK, B0K1, B0M1, B0N1 B0H3, B0M3, B1H1, B1H3, B1HA, B1HK, B1HU, B1K1, B1M1, B1M3, B1N1	<8>
18*i*-B	BDH1, BDH5, BDHA, BDHE, BDHQ, BDK1, BDM1, BDM5, BDN1 BDH3, BDM3, BEH1, BEH3, BEHA, BEHK, BEHU, BEK1, BEM1, BEM3, BEN1	
21*i*-B	DDH1, DDHA, DDHK, DDK1 DEH1, DEHA, DEHK, DEHU, DEK1	<8>

5 LOG OUTPUT

5.1 LOG OUTPUT

The result of error check and writing is output to the following folder. <2>
<C:\ProgramData\FANUC\CNCDataManagementTool\Log>

This folder is hidden by default. Refer to "Settings after installation" in "2.2.1 INSTALL" for the reference method. <2>

The writing result is output to the log folder. The output file name will be "yyyymmdd.log". “yyyymmddHHMMSS” in the file name and output example is the date and time when the log was output.

- yyyy : year
- mm : month
- dd : day
- HH : hour
- MM : minute
- SS : second

Output example)

(1) Start and end of writing (when writing to 30i-B)
- yyyy/mm/dd HH:MM:SS Write Start Series:[30i-B(Gxxx/aaaa)] ID:[00000000.00000000.00000000.00000000] IP:[127.0.0.1] <4>
- yyyy/mm/dd HH:MM:SS Write End Series:[30i-B(Gxxx/aaaa)] ID:[00000000.00000000.00000000.00000000] IP:[127.0.0.1] <4>

"xxx" shows the series name, "aaaa" shows the version number, and “ID” shows the CNC ID information. <4>

(2) Files written successfully
- yyyy/mm/dd HH:MM:SS Success Section:[ToolOffsetData] File:[C:\Data\TOOLOFST.TXT]

(3) File for which writing ended in error
- yyyy/mm/dd HH:MM:SS Error Section:[ToolOffsetData] File:[C:\Data\TOOLOFST.TXT] errorcode:[1000] detail:[0/0/0]

(4) When writing of the program file is stopped by the registered program <6>
- yyyy/mm/dd HH:MM:SS Stop Section:[Program] File:[C:\Data\O100] Path:[//CNC_MEM/USER/PATH1/] Prog:[O0100]

(5) When writing of the registered program is skipped in writing the program file
- yyyy/mm/dd HH:MM:SS Skip Section:[Program] File:[C:\Data\O100] Path:[//CNC_MEM/USER/PATH1/] Prog:[O0100] <6>

(6) When there is a block that cannot be determined in the check process for writing the program file (The following log and “(3) File for which writing ended in error” log are output.) <7>
- yyyy/mm/dd HH:MM:SS Multiple block program numbers / names. Error block: [O0001O0002O0003] <7>

(7) If the file specified in the “definition file” does not exist
- yyyy/mm/dd HH:MM:SS Error File is not found. Section:[ToolOffsetData] File:[C:\Data\TOOLOFST.TXT]

(8) If a file that is not specified in the “definition file” exists in the “storage folder”
- yyyy/mm/dd HH:MM:SS Error File is not defined. File:[TOOLOFST.TXT]

(9) Notification of the “definition file” to be used <2>
- yyyy/mm/dd HH:MM:SS Information Definition File:[C:\Data\Sample.ini] <2>

(10) Communication error occurred <5>
- yyyy/mm/dd HH:MM:SS Connect error. IP:[127.0.0.1] PORT:[8193] Error code:[1023] <5>

*) Outputs of (7) and (8) are output regardless of the setting of "Option" of "3.4 SETTING". However, if "Don't make an error" is selected, a warning will be output instead of an error.

How to check the error code <2>

If the writing of data to the CNC ends abnormally, the following will be output to the log.
- yyyy/mm/dd HH:MM:SS Error Section:[ToolOffsetData] File:[C:\Data\TOOLOFST.TXT] errorcode:[XXXX] detail:[0/0/YY]

XXXX and YY are the error code and detailed code in "Table 6.2 (a) Error code list". <2>

6 ERROR CHECKING

6.1 ERROR CONTENTS

Click the [Show list] / [Execution] button on the startup screen (Fig. 3.1 (a)) to execute the error check. <5>

No.	Description	Supplement
1	The “storage folder” selected on the startup screen (Fig. 3.1 (a)) does not exist	
2	When multiple “definition file” are in the “storage folder” without selecting the “definition file” on the startup screen (Fig. 3.1 (a))	
3	When a “definition file” is selected on the startup screen (Fig. 3.1 (a)) and the “definition file” does not exist	
4	The file specified in the “definition file” does not exist	*1
5	A file not specified in the “definition file” exists in the “storage folder”	*1
6	Definition file "Folder key" is not specified	
7	Definition file "Path key" is not specified	
8	There is no file to write	
9	Unable to open the target file (for example, editing with another editor)	

* 1) If the toolbar option is set to no error, no error is output. The corresponding file name is printed in the log.

6.2 ERROR CODE LIST <2>

The error codes output to messages and logs during communication between the CNC and the “CNC Data Management Tool” are as follows. If an error occurs, check "Explanation and Error handling". If you receive an error code not listed, please contact FANUC.

Table 6.2(a) List of error codes

Error code	Meaning	Explanation and troubleshooting actions (The numbers in () are detailed codes)
1004	Specified function error	It may not be a compatible model. Refer to "CORRESPONDENCE BY MODEL" and confirm that it is a write target.
1005	Data error	Please check the data and correct it. If the details of the error are output, the following causes are possible. - When writing a TXT file (excluding programs) (1)A character which is unavailable for NC program is detected. - When writing a program file (1)A character which is unavailable for NC program is detected. (2)When TV check is effective, a block which includes odd number of characters (including 'LF' at the end of the block) is detected. (3)The registered program count is full. (4)The same program number has already been registered. (5)The same program number is selected on CNC. - When writing an application file (1)F-ROM writing failed (defective device) (3)F-ROM erasing failed (defective device) (6)No writing space in F-ROM (7)The F-ROM data not permitted at the CNC system was specified. (8)The size of the system data actually written on the F-ROM and the system data being specified is different. (9)F-ROM(MODULE) cannot be recognized. (defective device) (144) If it occurs when reading option information file, alarm PS5523 “OPTION AUTHENTICATION WAIT STATE” has occurred. (145)The total data size is excessed.
1006	No option	Refer to "CORRESPONDENCE BY MODEL" and confirm that it is a write target.
1007	Write protection	Write operation is prohibited.

<2> <2> <4> <4> <2> <2> <7> <7> <2> <2>

Error code	Meaning	Explanation and troubleshooting actions (The numbers in () are detailed codes)
1008	Memory overflow	CNC tape memory is overflowed.
1009	CNC parameter error	CNC parameter is set incorrectly.
1011	Path number error	A path number is incorrect.
1012	CNC mode error	The CNC mode is incorrect. Correct the CNC mode.
1013	CNC execution rejection	The execution at the CNC is rejected. Check the condition of execution.
1014	Alarm	The function cannot be executed due to an alarm in CNC. Remove the cause of alarm.
1016	State of data protection	Data is protected by the CNC data protection function.
1018	Reset or stop request	The RESET or STOP button was pressed. Call the termination function.
1023	Socket error	Investigate CNC power supply, Ethernet cable and I/F board. Check "2.3.1 Settings for Tool Side" and "2.3.2 Settings for CNC Side", and restart the CNC and tool.
2001	No PMC	The PMC does not exist.
2005	Data error	Please check the data and correct it. If the details of the error are output, the following causes are possible. (1)The first character string was not "%;". (2)The end of the character string was not ";(0x0A)" or "%". (3)The character string of second block was not the header information. (4)A model name of header information and an actual model are different. (5)The character by which header information and ID information were delimited was not ",". (6)The character string of ID information on header information was not "MSID =". (7)The MSID number on header information and current PMC unit type are different. (8)The last character of header information was not ")". (9)The first character of parameter block was not "N". (10)The address number of parameter block was wrong. (no six digits) (11)The address number of parameter block was wrong. (the head of N number is not "6") (12)The third character of the parameter block was not "0", "1", "2", "3", "4", or "9". (13)The block of larger group information than the total of the group was specified. (14)The data table control information was wrong. (number which cannot be specified) (15)The head of the character for the set value on the parameter block was not "P". (16)The character which was not the numerical value was included in the setting value. (format error of setting value) (17)The number of characters of bit patterns was not eight characters. (format error of setting value) (18)The range of address was wrong. (19)The total number of data table control information was wrong. (20)The data table control information was wrong.
2006	No option	There is no corresponding CNC option.
2008	State of data protection	Data is protected by the CNC data protection function.
2009	Abnormal library state	If this occurs in writing PMC(DCS) parameters, parameter setting may be impossible (because bit 0(PWE) of parameter No.8900 is 0).
2013	Socket error	Investigate CNC power supply, Ethernet cable and I/F board. Check "2.3.1 Settings for Tool Side" and "2.3.2 Settings for CNC Side", and restart the CNC and tool.
2999	Other errors	If this occurs in writing PMC(DCS) parameters, parameter setting may be impossible (because bit 0(PWE) of parameter No.8900 is 0).
3001	File open error	The file may not exist or may be opened by another process.

Error code	Meaning	Explanation and troubleshooting actions (The numbers in () are detailed codes)
3002	Illegal write data	Automatic data backup data to be written is not the file read by this tool. Only the file read by this tool can be written.
3003	Illegal write program	This is a program file that cannot be written by this tool. If the error details are output, the following causes are possible. (1)There are multiple program number or program name in one block. (2)The program number or program name is incorrect In case of (1), take one of the following actions. - Correct the block of program output in the log - Empty the file in the target folder to write - On the option setting screen of the tool, set the operation when 2(REP) of parameter No.3201 is 1 and the same program is registered to "Overwrite all". In case of (2), take the following actions. - Correct the block of program output in the log
3004	Not writable	Unable to write to this file due to the version of CNC software.
3005	Illegal write workpiece coordinate system shift	Please check the data and correct it. If the details of the error are output, the following causes are possible. (1) Wrong number of digits in write data(10 digits or more) (2) Axis name does not exist in target CNC (3) Wrong data (missing decimal point, illegal characters, etc.)
3006	Illegal write software operator's panel	Please check the data and correct it. If the details of the error are output, the following causes are possible. (1) Wrong value (DATA) (DATA is not separated by TAB code, there is no data, etc.)
3007	Illegal write software operator's panel general-purpose switch	Please check the data and correct it. If the details of the error are output, the following causes are possible. (1) Target switch number (NO) is wrong (other than 1 to 16) (2) Wrong switch name (NAME) (9 characters (9 bytes) or more) (3) Wrong value (DATA) (Other than 0/1)
3008	Wrong write to P-code variable	Please check the data and correct it. Possible causes are: - 16i/18i/21i-MODEL B dedicated format does not exist or is invalid (when target is 16i/18i/21i-MODEL B) - When data in parentheses after G10LxxPxxx is invalid (data other than 16 digits such as '0' to '9' and 'A' to 'F')
3009	Unable to read operation history	For 16i/18i/21i-MODEL B, one of the following parameters is set to "1". - No.3112#5 - No.8700#0

【Version history】

Ed	Date	Description
01	May.29, 2020	The first edition registration
02	Jun.23, 2020	Add or Change at <2> part
03	Sep.14, 2020	Add or Change at <3> part
04	Nov.04, 2020	Add or Change at <4> part
05	Feb.17, 2021	Add or Change at <5> part
06	Jun.18, 2021	Add or Change at <6> part
07	Dec.01, 2021	Add or Change at <7> part
08	Mar.31, 2022	Add or Change at <8> part
09	Jul.12, 2022	Add or Change at <9> part
10	Oct.24, 2022	Add or Change at <10> part
11	Aug.31, 2023	Add or Change at <11> part
12	Dec.20, 2023	Add or Change at <12> part
13	Jul.12, 2024	Add or Change at <13> part
14	Jan.17, 2025	Add or Change at <14> part